

BFS Traversal

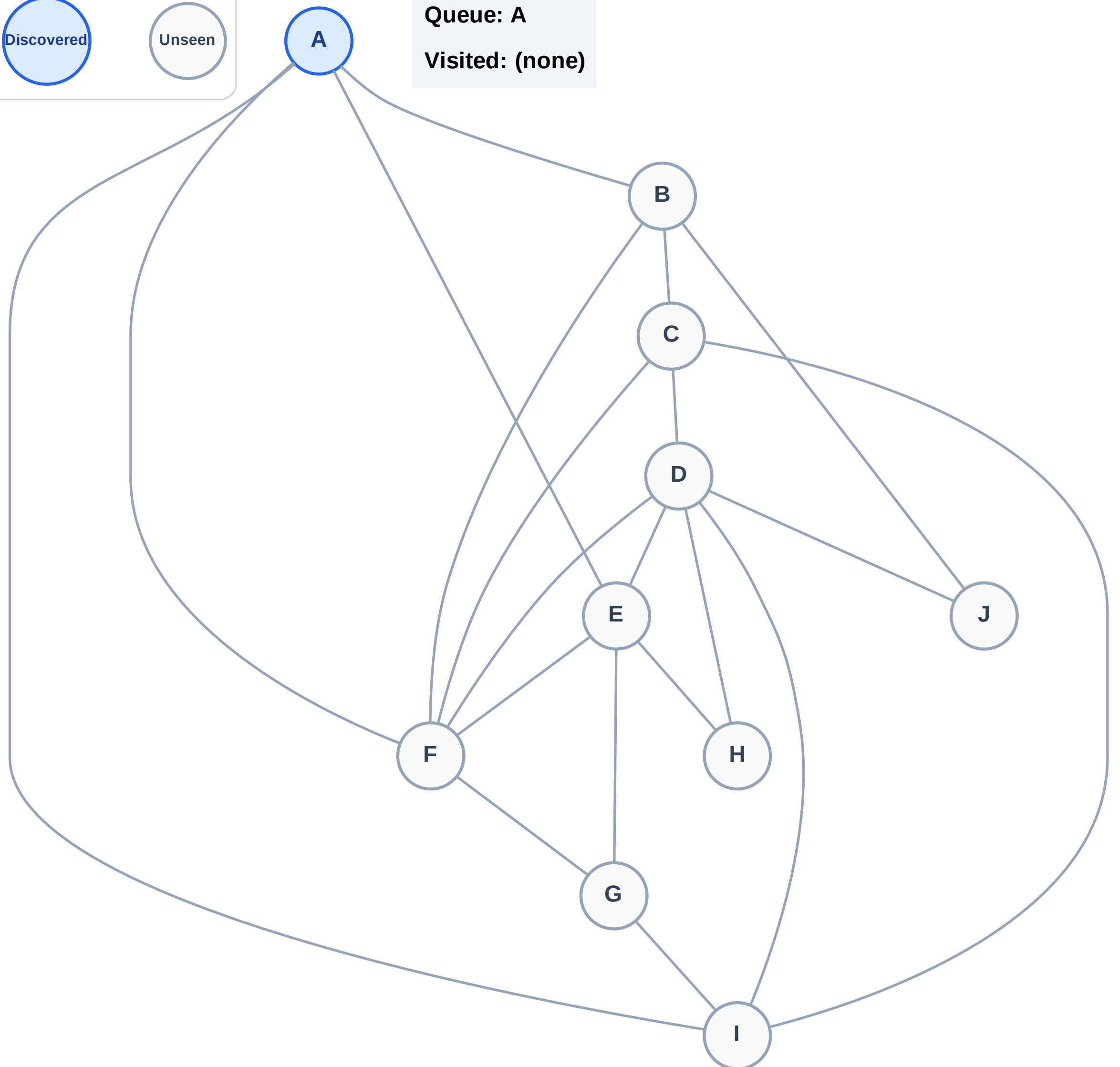
Step 1: Start at A

Legend



Queue: A

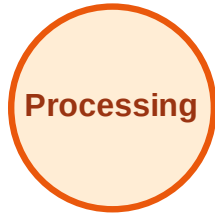
Visited: (none)



BFS Traversal

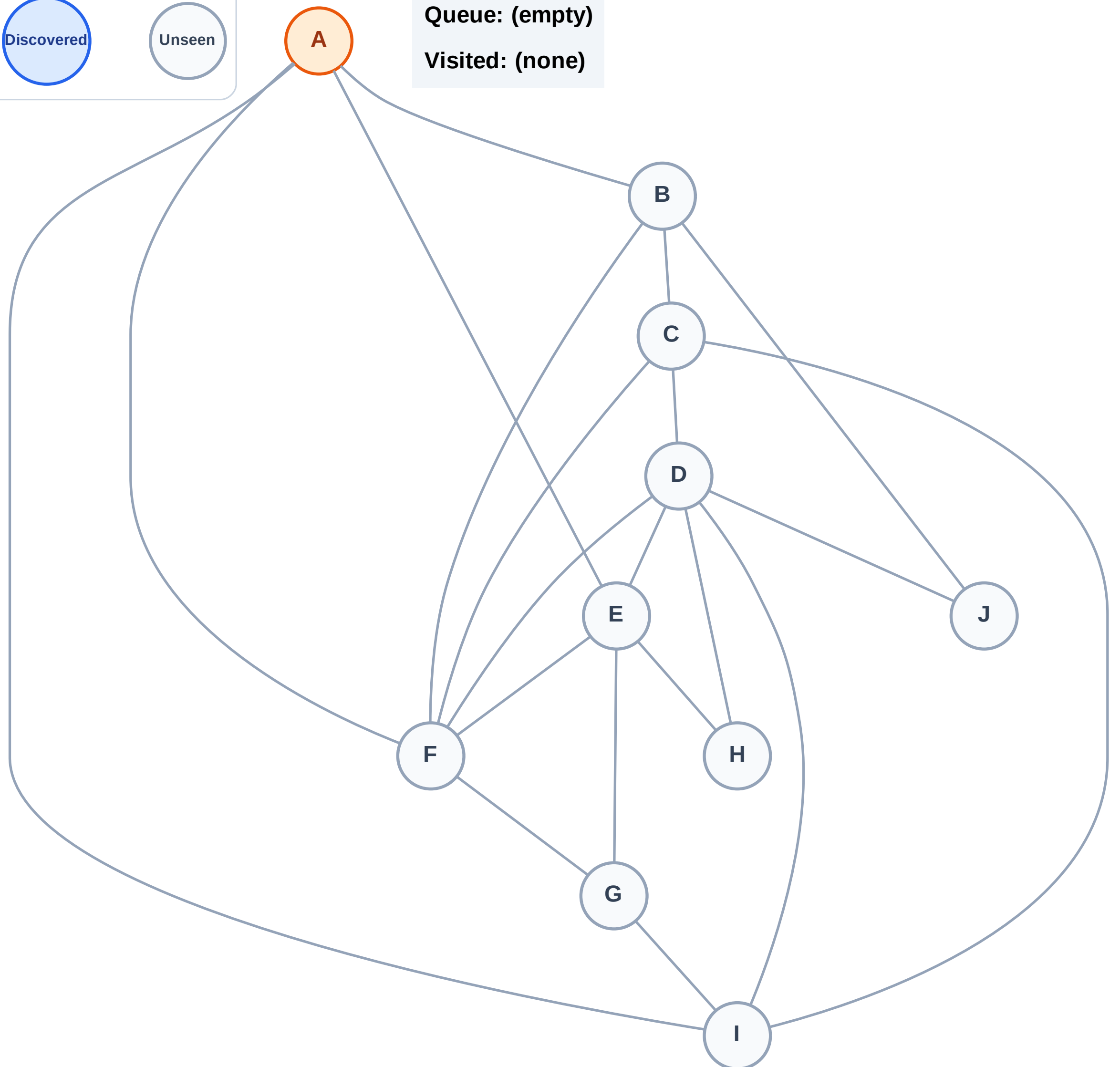
Step 2: Process node A

Legend



Queue: (empty)

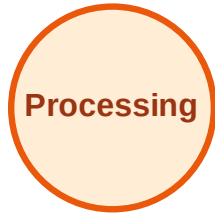
Visited: (none)



BFS Traversal

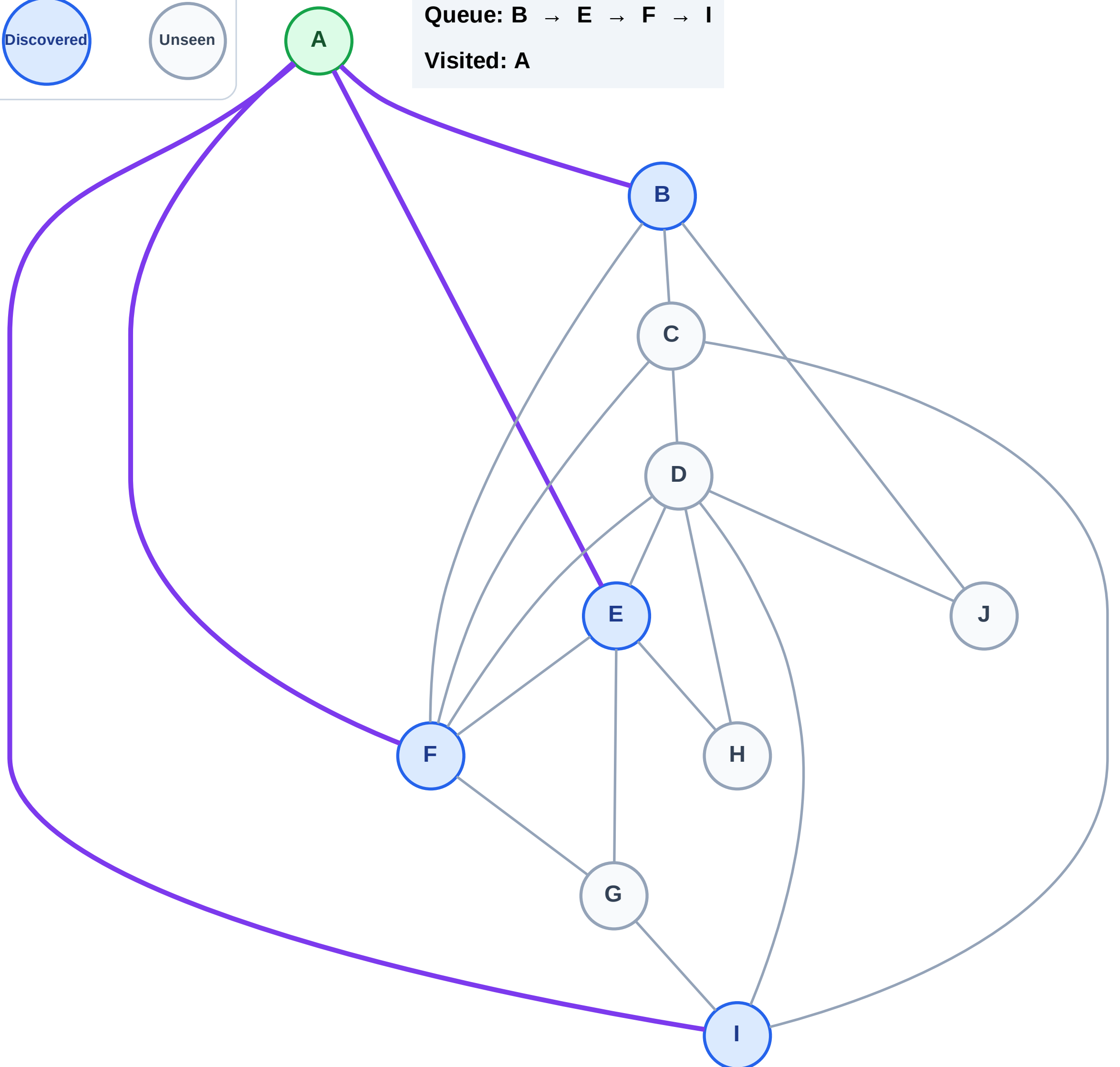
Step 3: Discover B, E, F, I from A

Legend



Queue: B → E → F → I

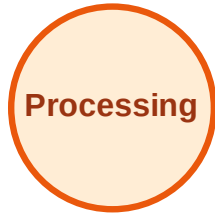
Visited: A



BFS Traversal

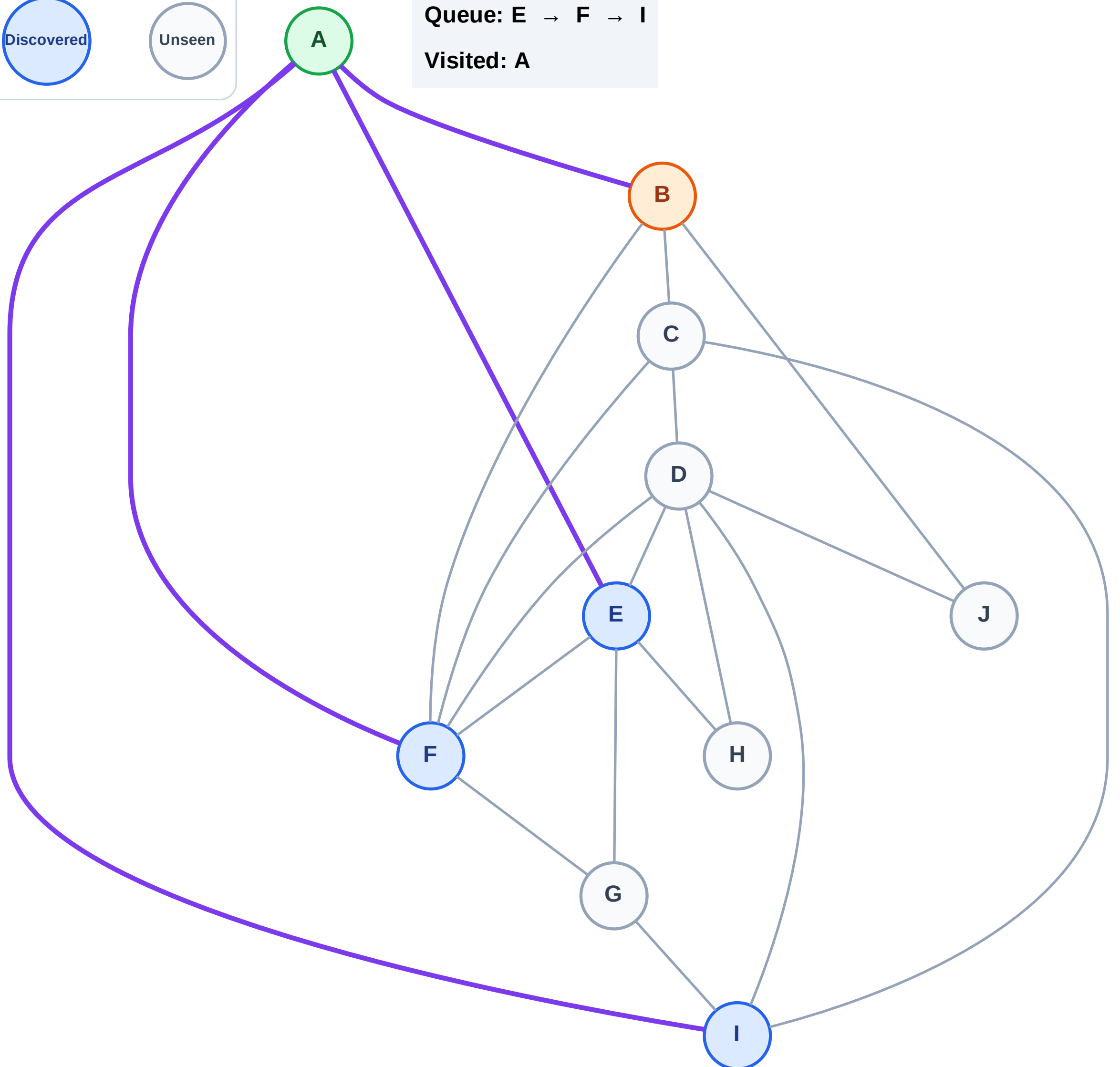
Step 4: Process node B

Legend



Queue: E → F → I

Visited: A



BFS Traversal

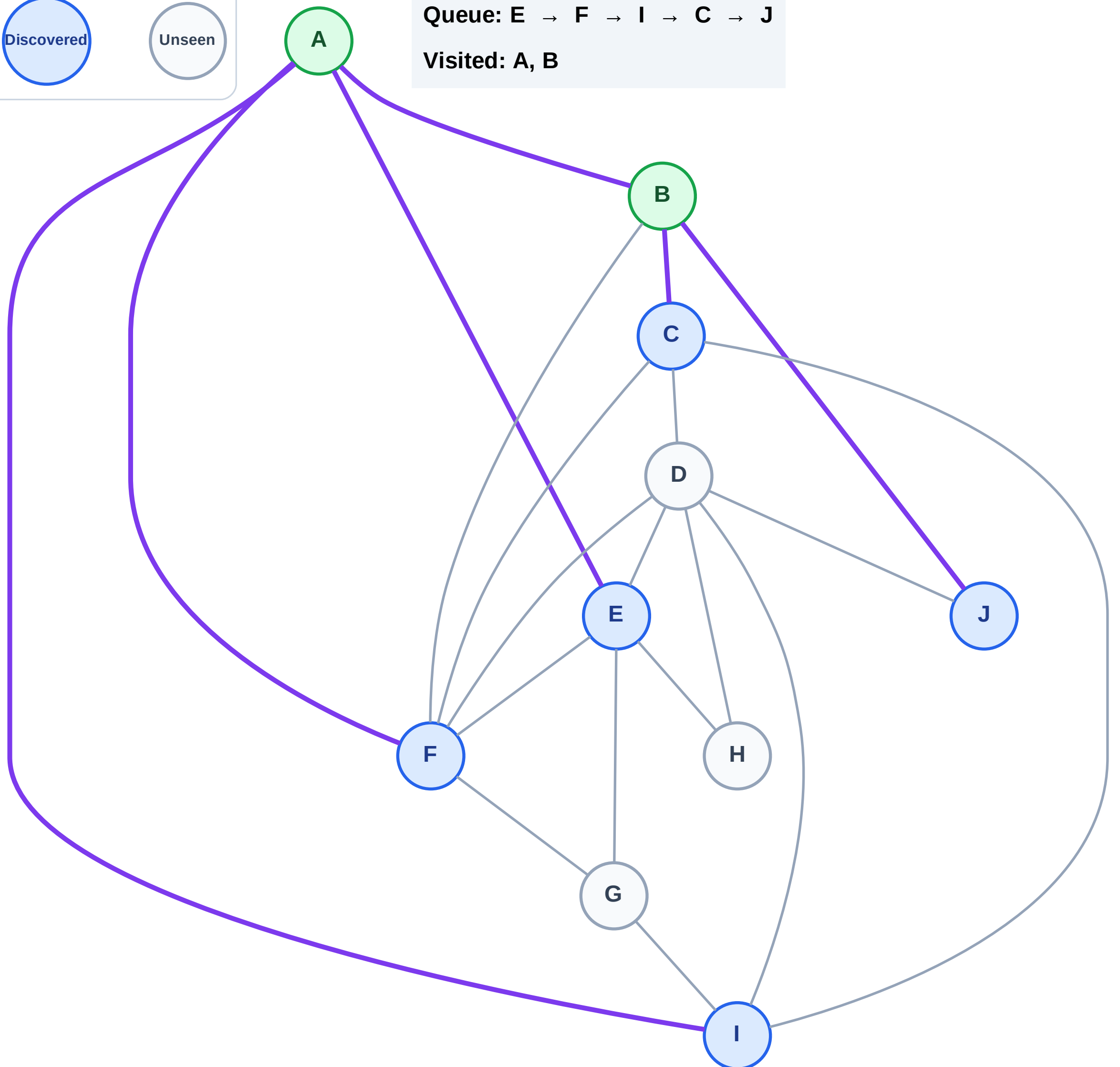
Step 5: Discover C, J from B

Legend



Queue: E → F → I → C → J

Visited: A, B



BFS Traversal

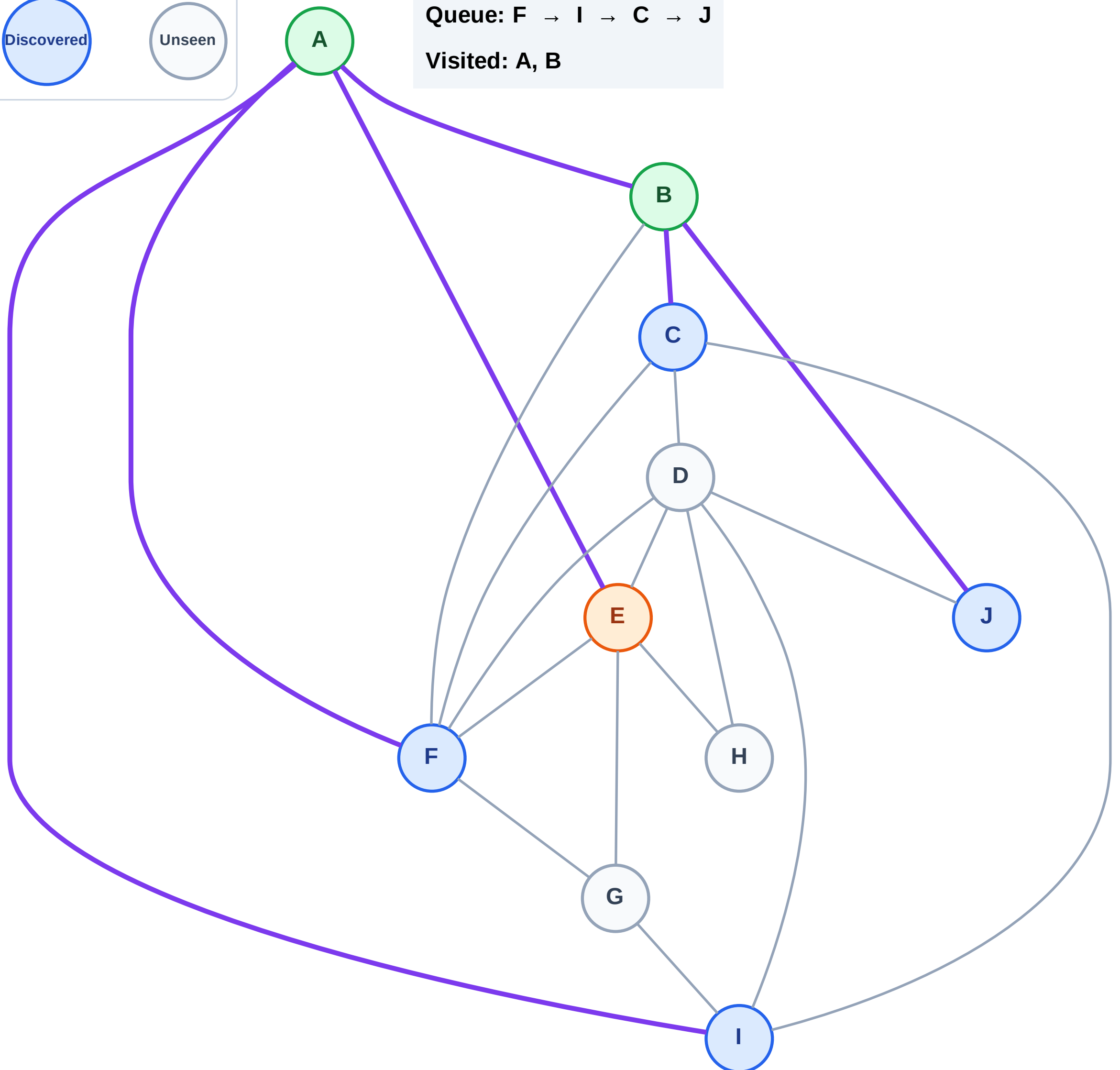
Step 6: Process node E

Legend



Queue: F → I → C → J

Visited: A, B



BFS Traversal

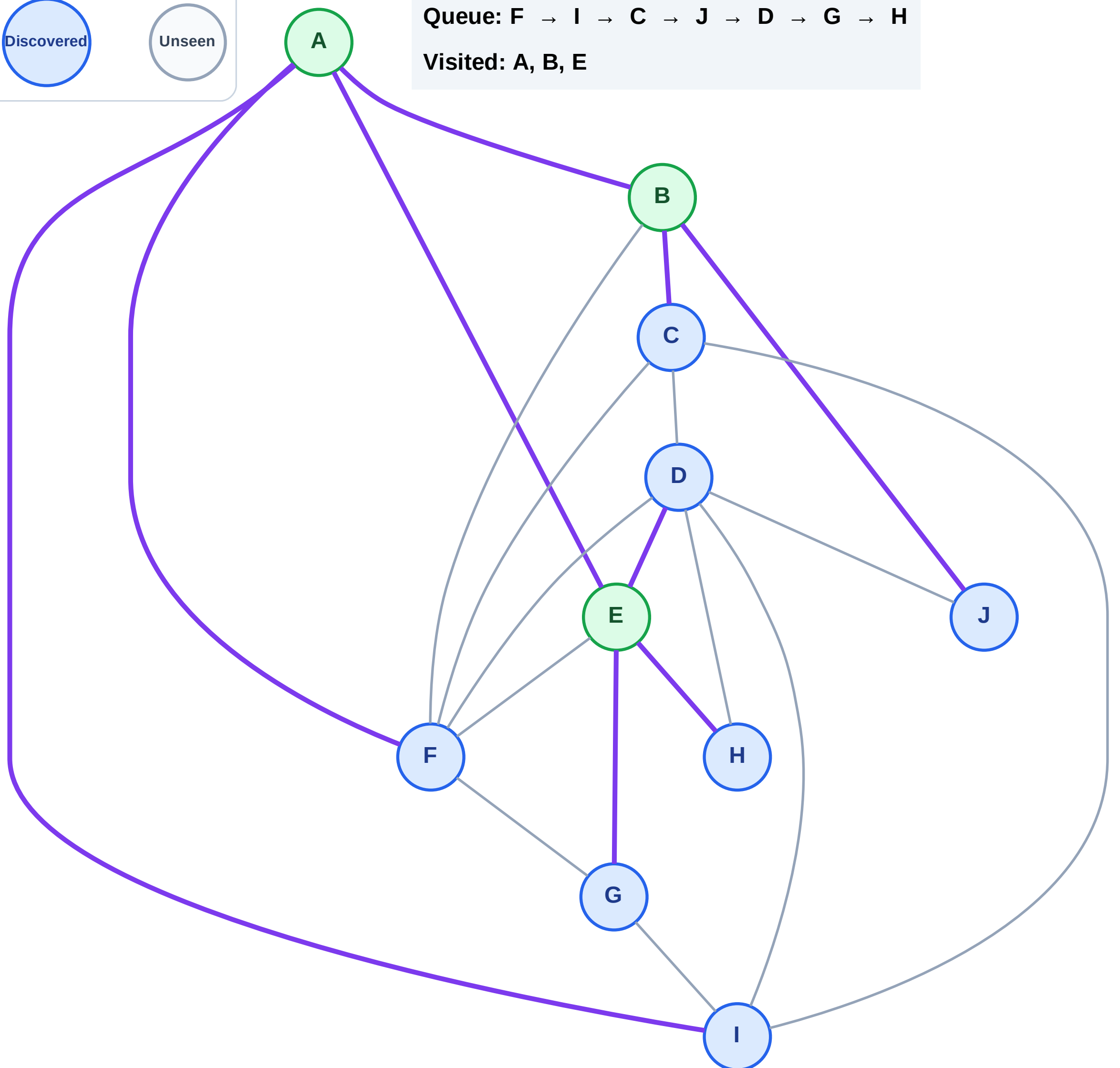
Step 7: Discover D, G, H from E

Legend



Queue: F → I → C → J → D → G → H

Visited: A, B, E



BFS Traversal

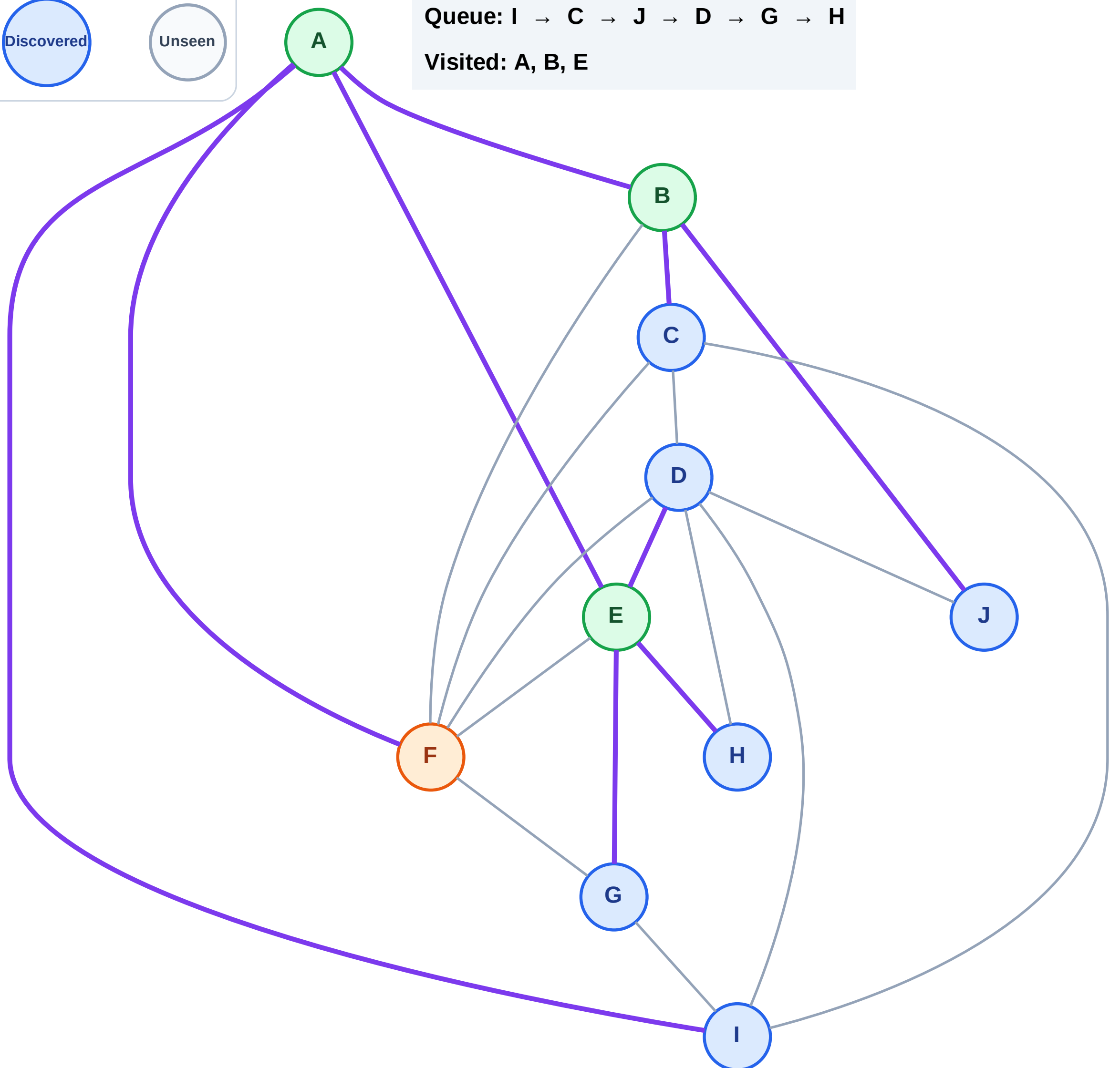
Step 8: Process node F

Legend



Queue: I → C → J → D → G → H

Visited: A, B, E



BFS Traversal

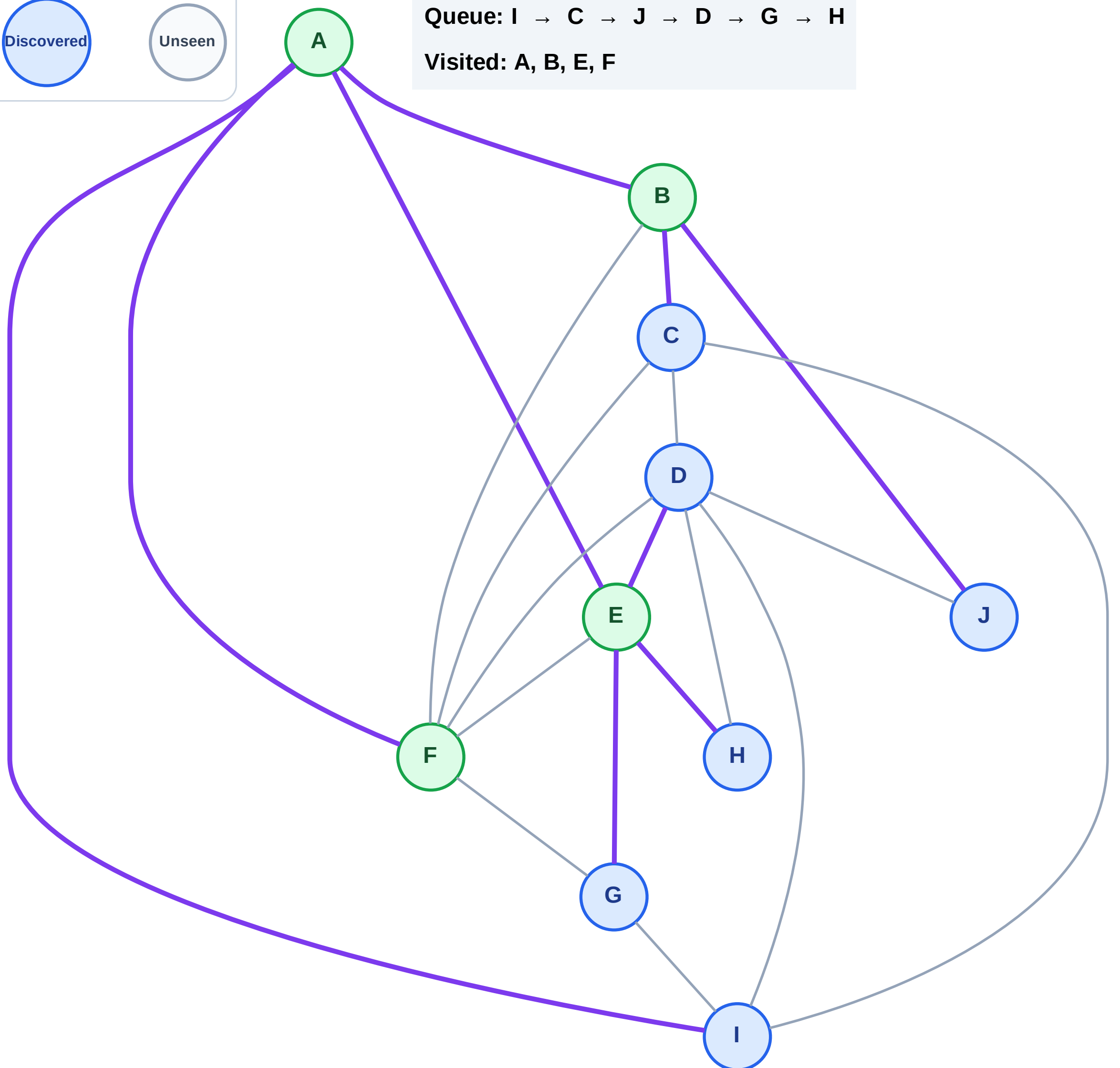
Step 9: No new neighbors from F

Legend



Queue: I → C → J → D → G → H

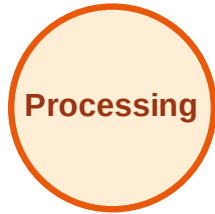
Visited: A, B, E, F



BFS Traversal

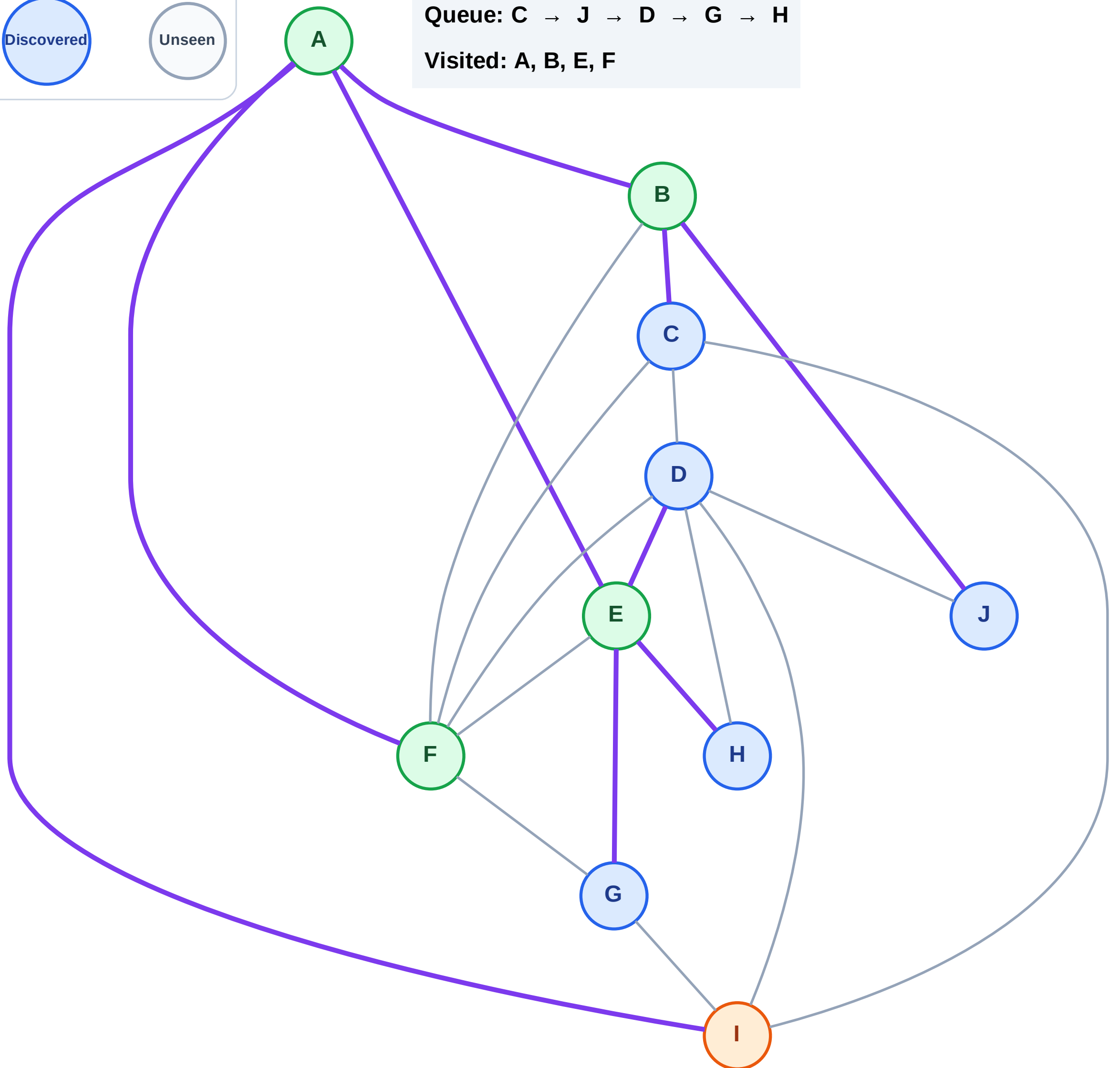
Step 10: Process node I

Legend



Queue: C → J → D → G → H

Visited: A, B, E, F



BFS Traversal

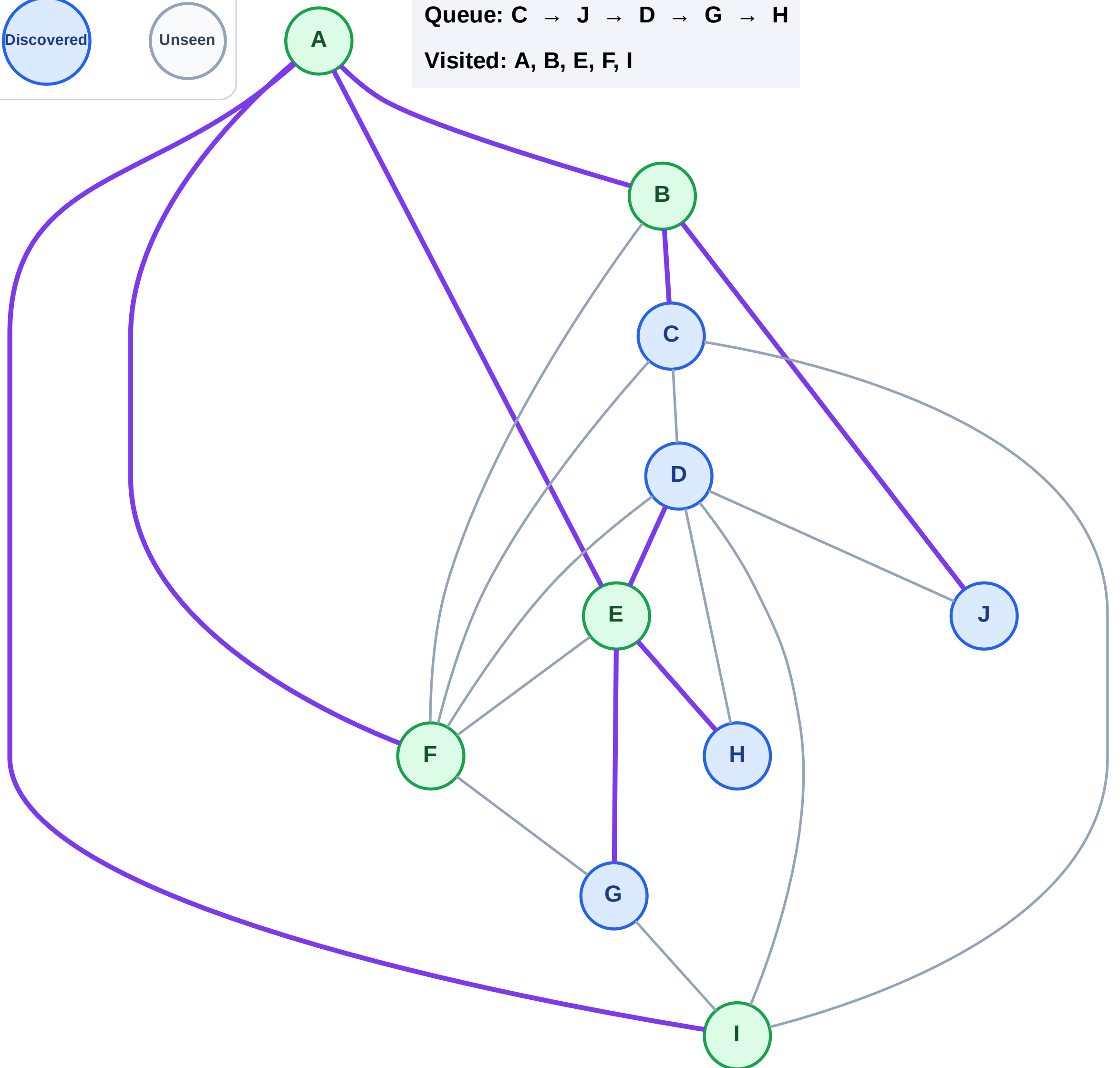
Step 11: No new neighbors from I

Legend



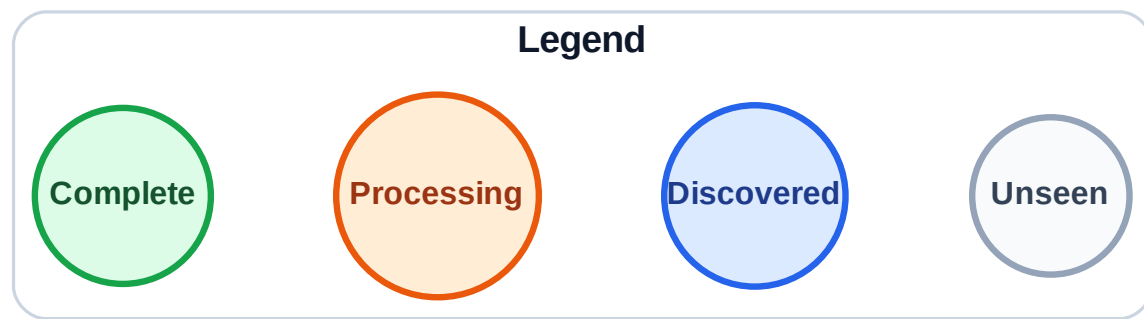
Queue: C → J → D → G → H

Visited: A, B, E, F, I

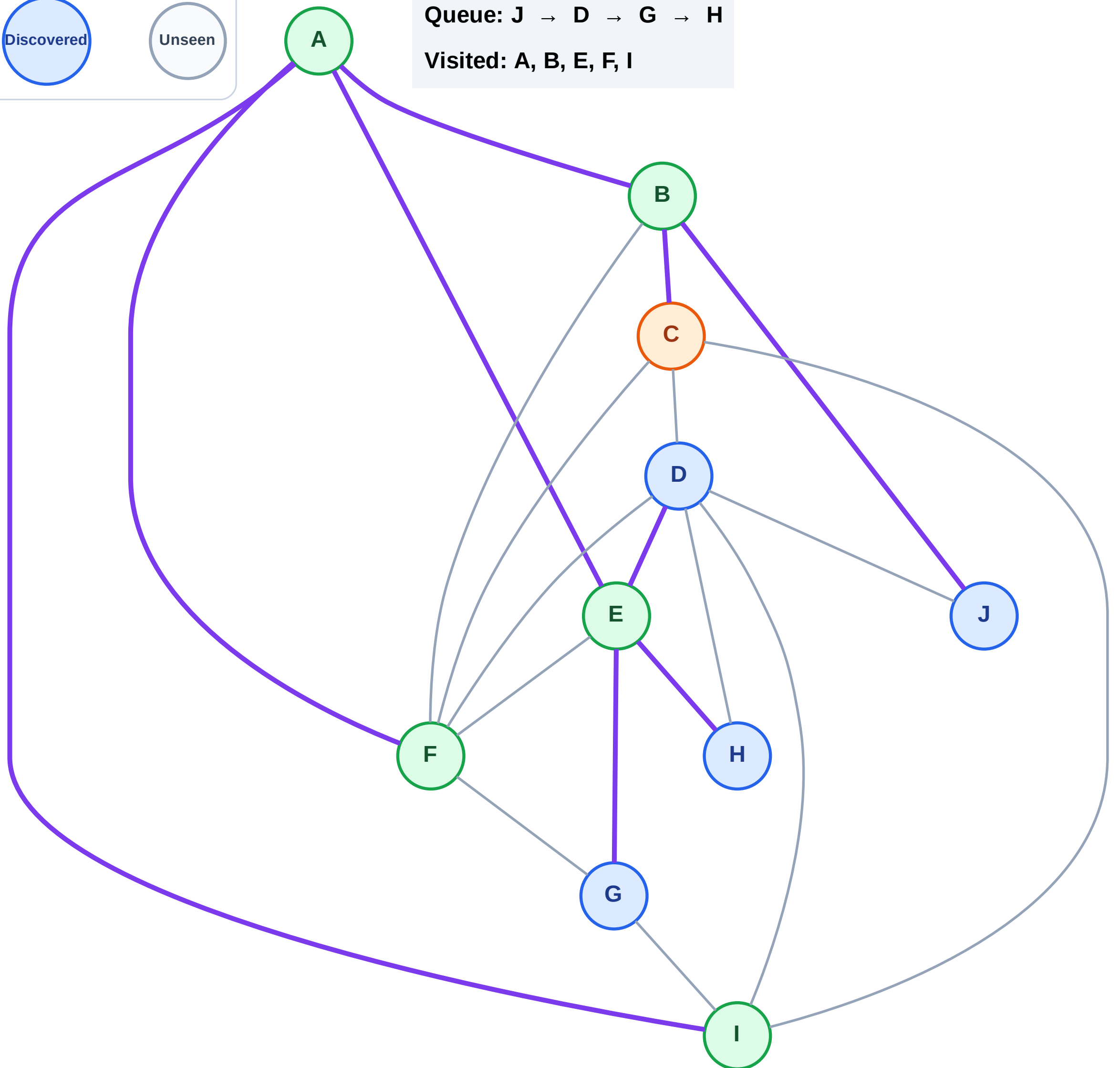


BFS Traversal

Step 12: Process node C



Queue: J → D → G → H
Visited: A, B, E, F, I



BFS Traversal

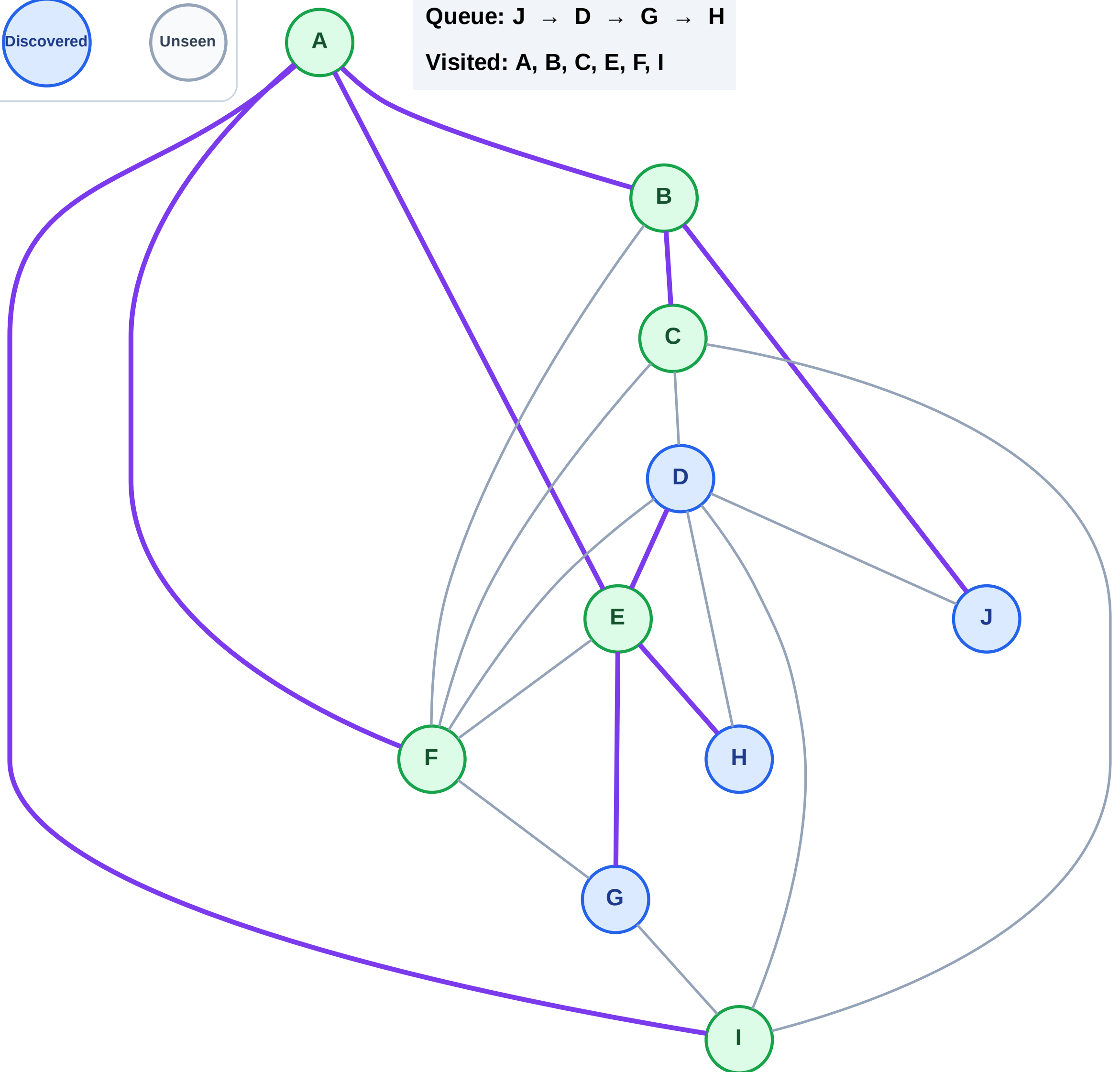
Step 13: No new neighbors from C

Legend



Queue: J → D → G → H

Visited: A, B, C, E, F, I



BFS Traversal

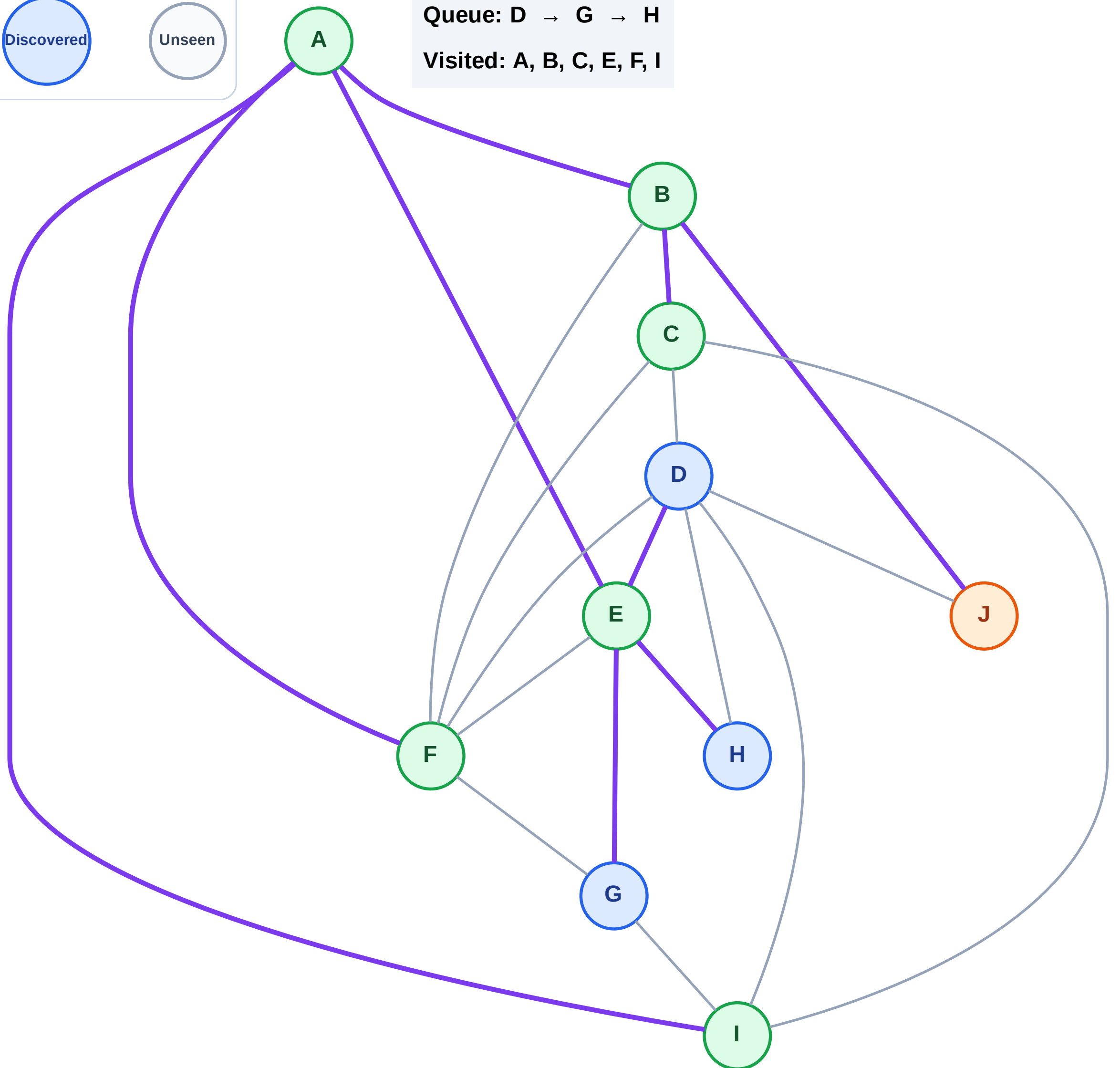
Step 14: Process node J

Legend



Queue: D → G → H

Visited: A, B, C, E, F, I



BFS Traversal

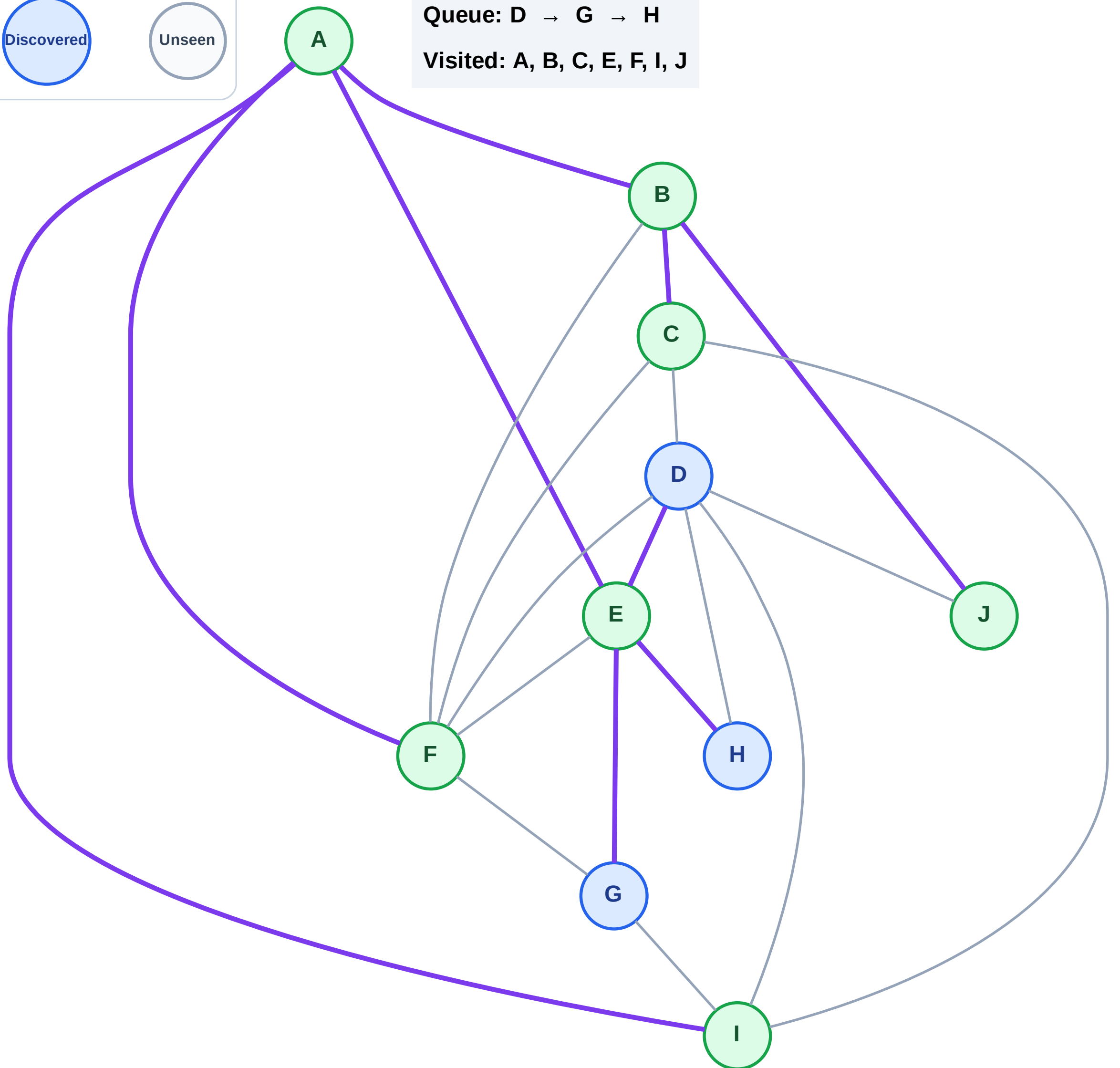
Step 15: No new neighbors from J

Legend



Queue: D → G → H

Visited: A, B, C, E, F, I, J



BFS Traversal

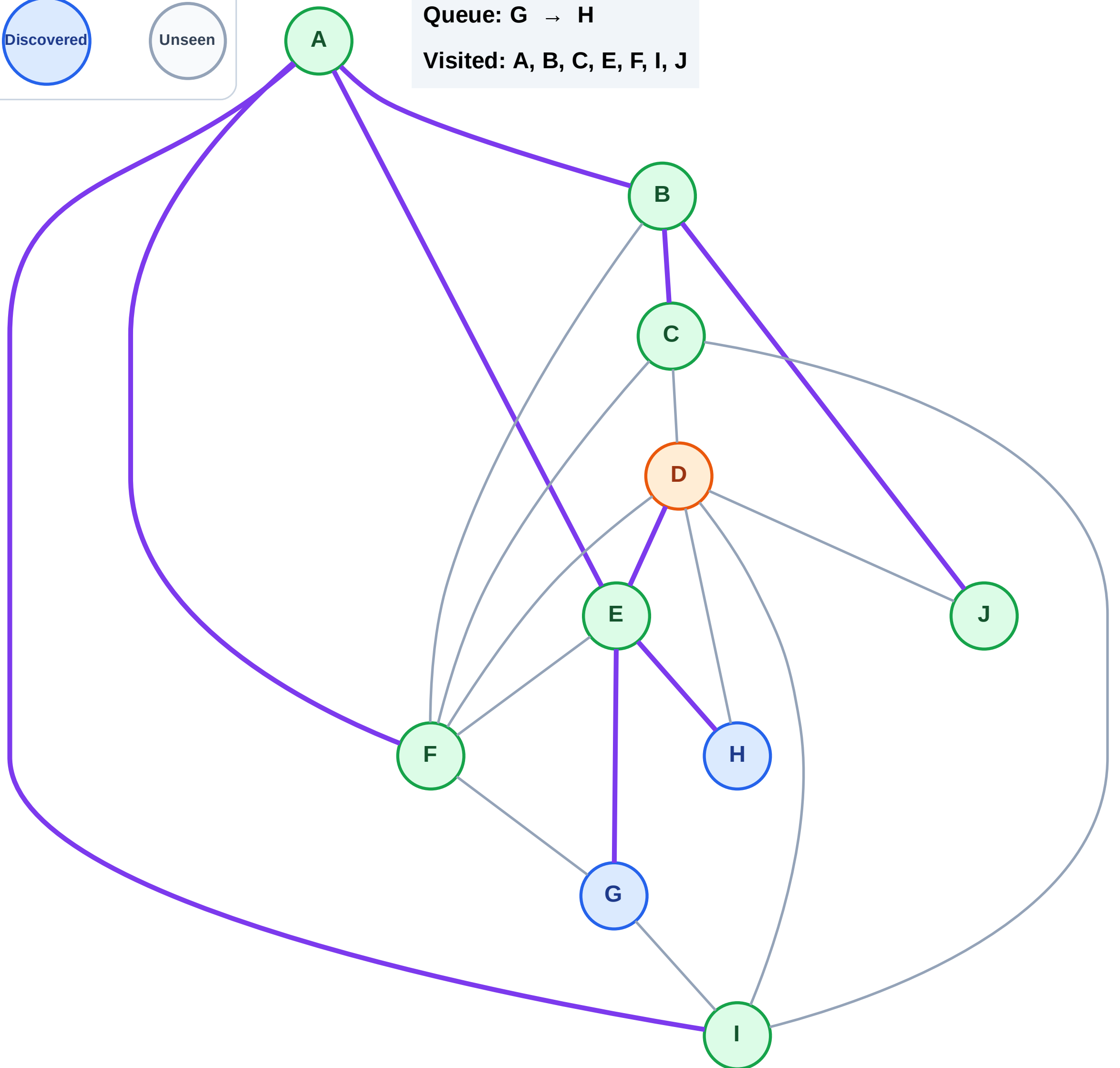
Step 16: Process node D

Legend



Queue: G → H

Visited: A, B, C, E, F, I, J



BFS Traversal

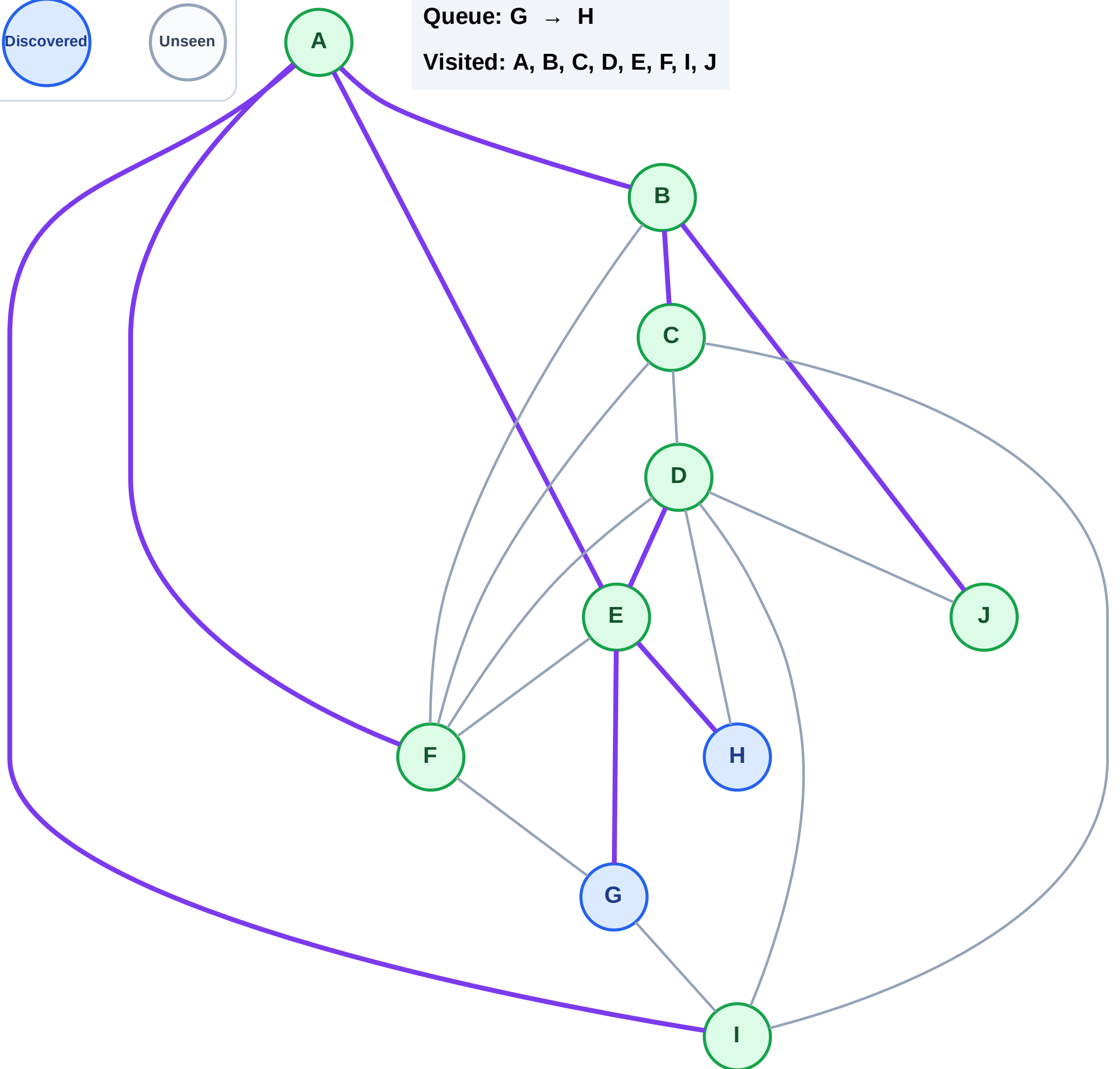
Step 17: No new neighbors from D

Legend



Queue: G → H

Visited: A, B, C, D, E, F, I, J



BFS Traversal

Step 18: Process node G

Legend

Complete

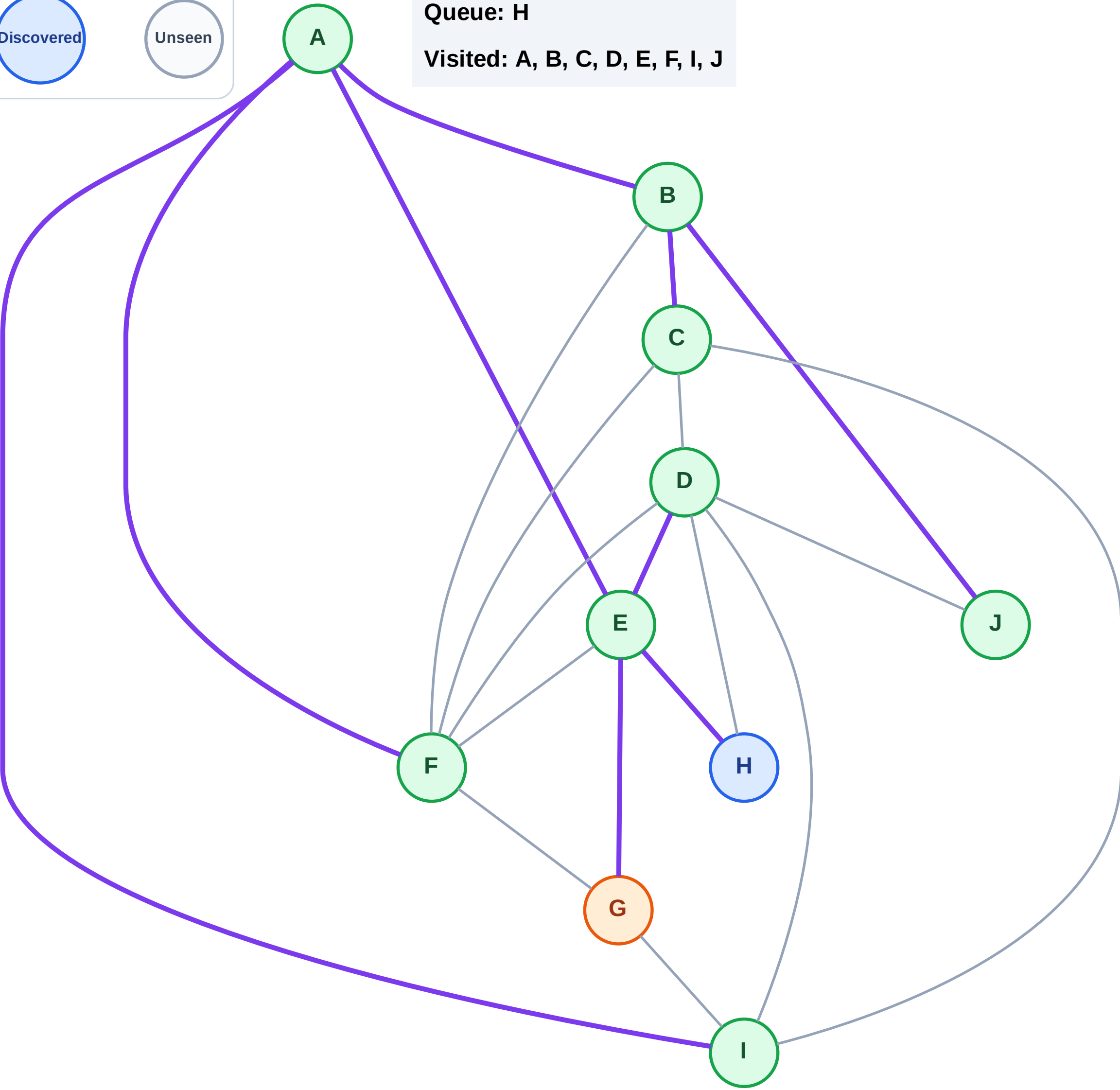
Processing

Discovered

Unseen

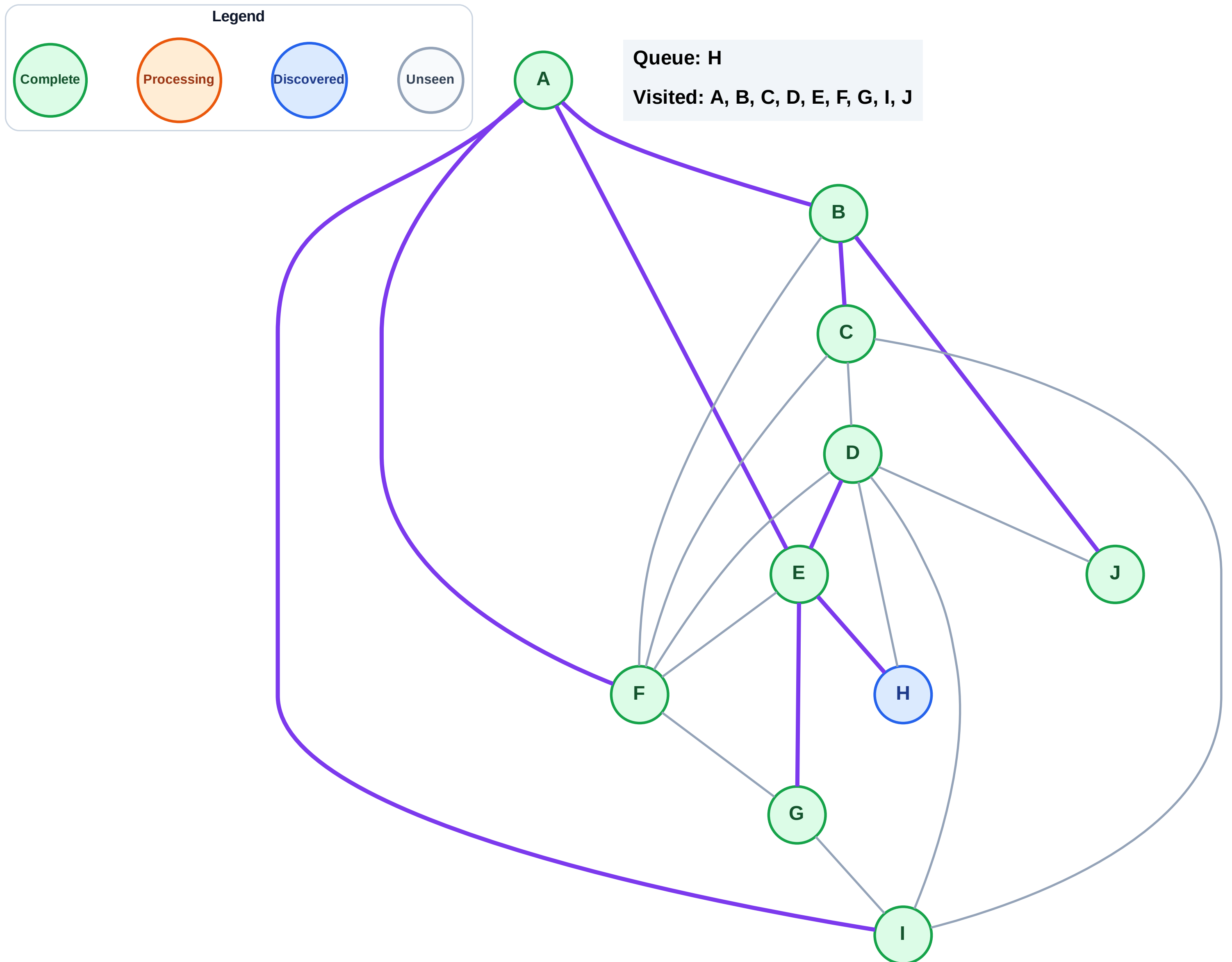
Queue: H

Visited: A, B, C, D, E, F, I, J



Step 19: No new neighbors from G

Step 19: No new neighbors from G



BFS Traversal

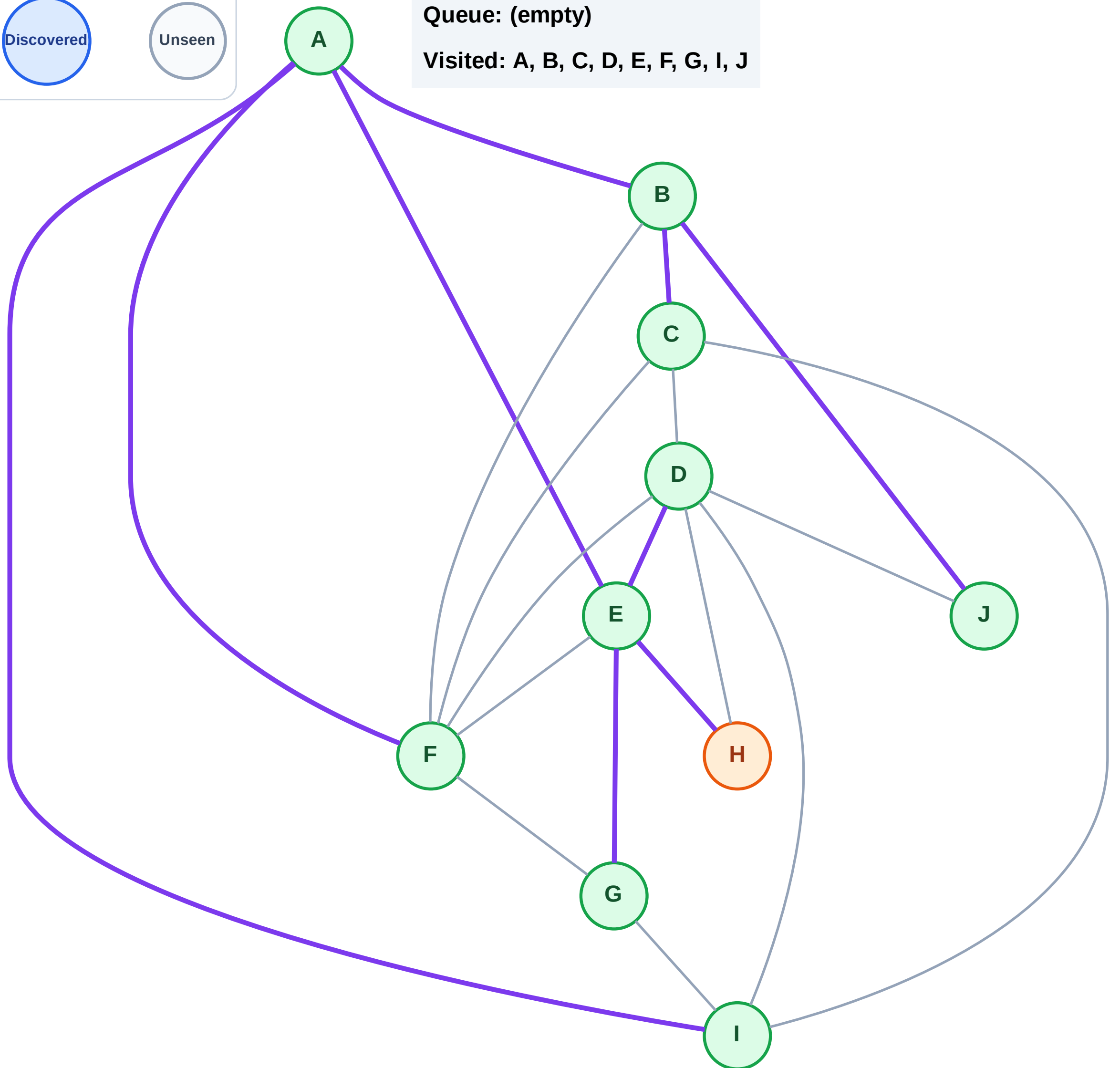
Step 20: Process node H

Legend



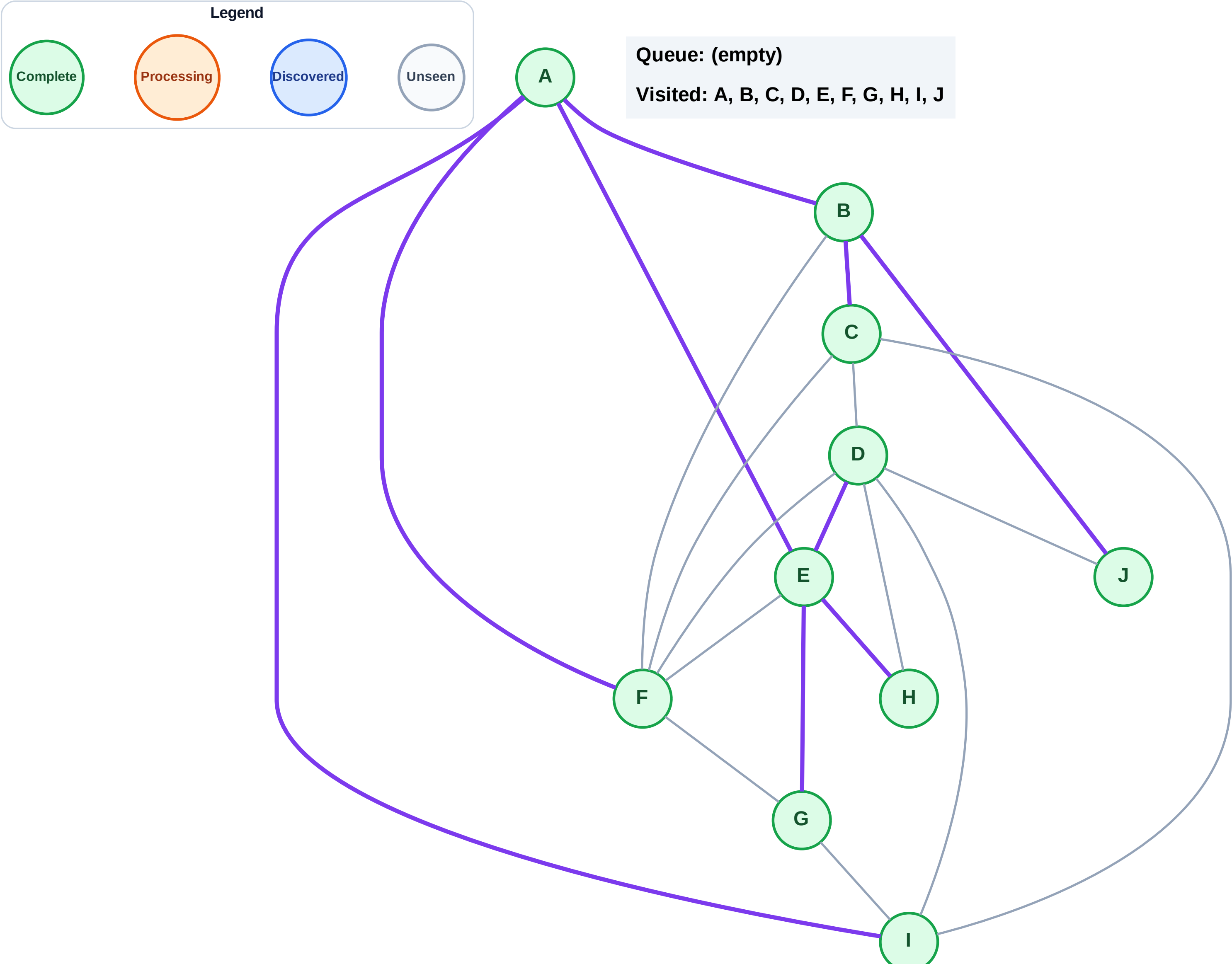
Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



BFS Traversal

Step 21: No new neighbors from H



DFS Traversal

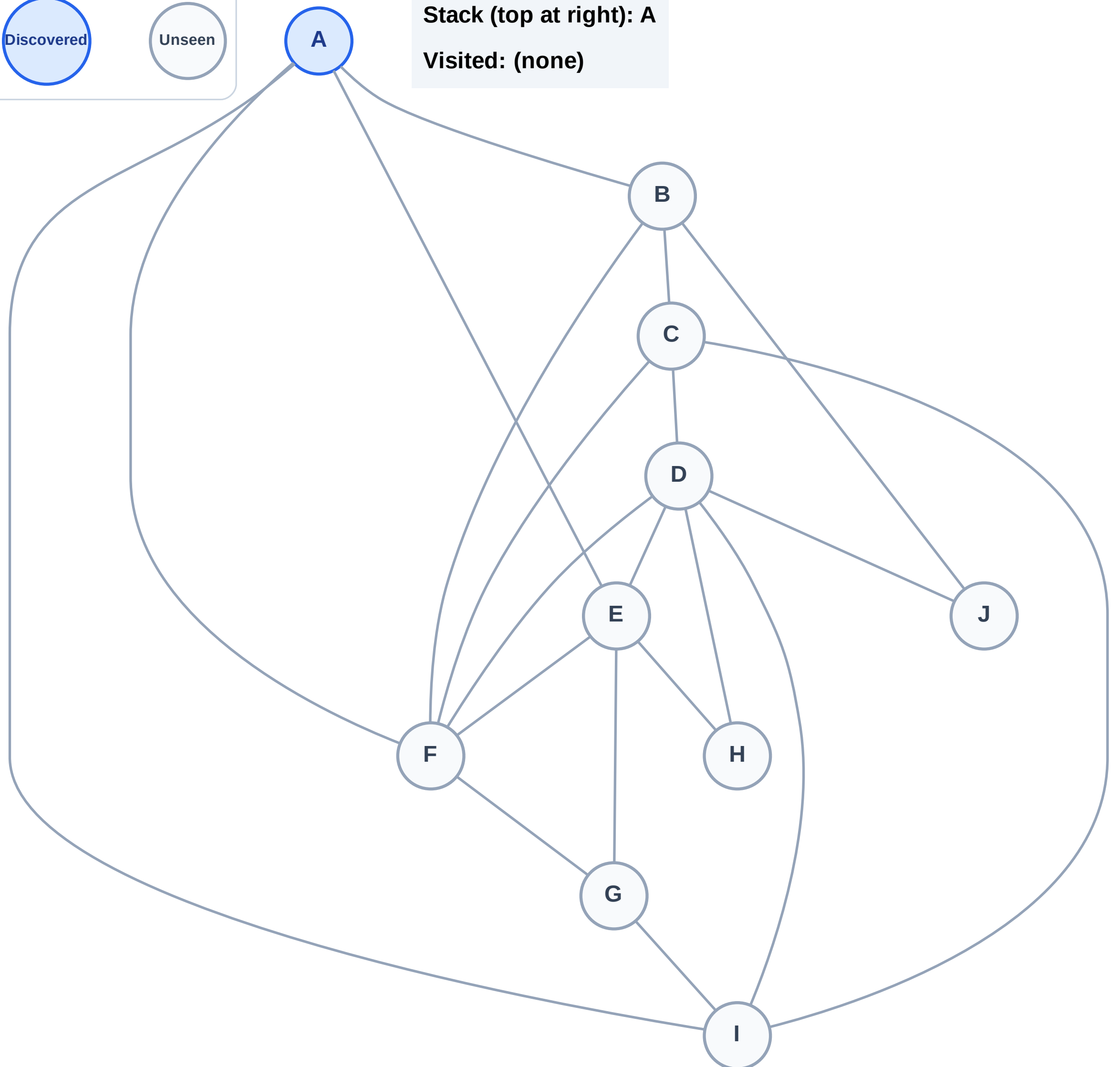
Step 1: Start at A

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

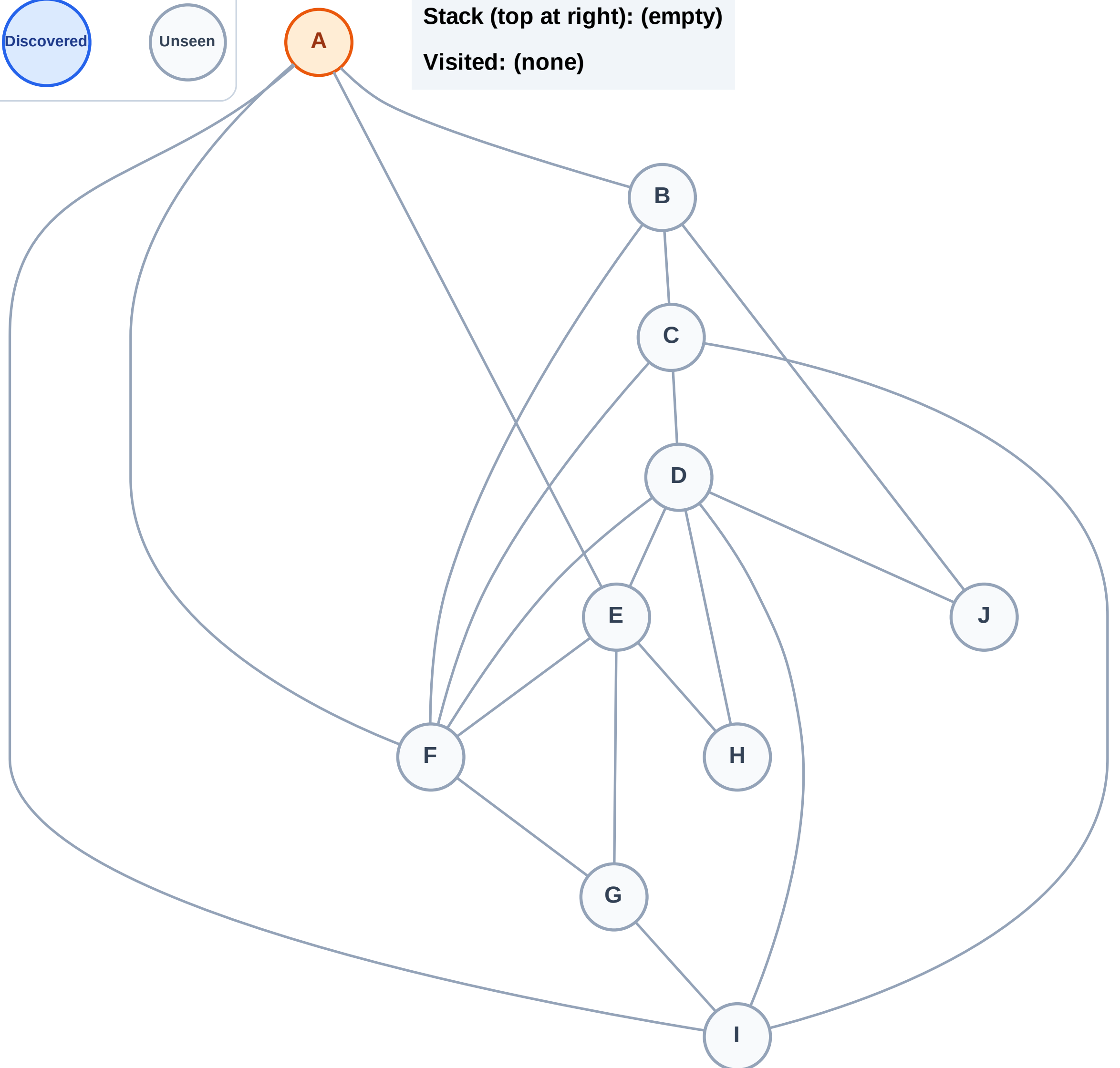
Step 2: Process node A

Legend



Stack (top at right): (empty)

Visited: (none)



DFS Traversal

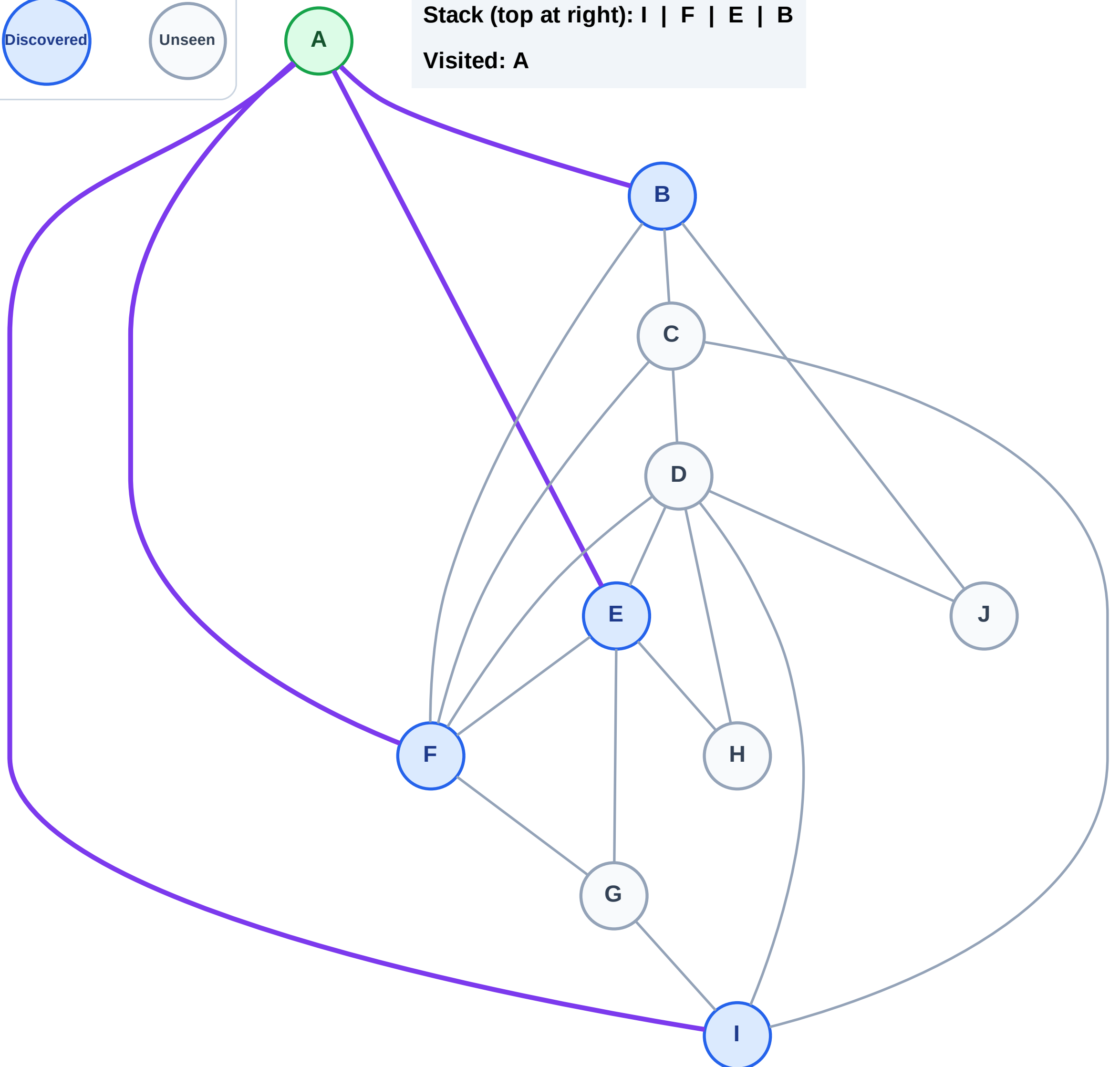
Step 3: Discover I, F, E, B from A

Legend



Stack (top at right): I | F | E | B

Visited: A



DFS Traversal

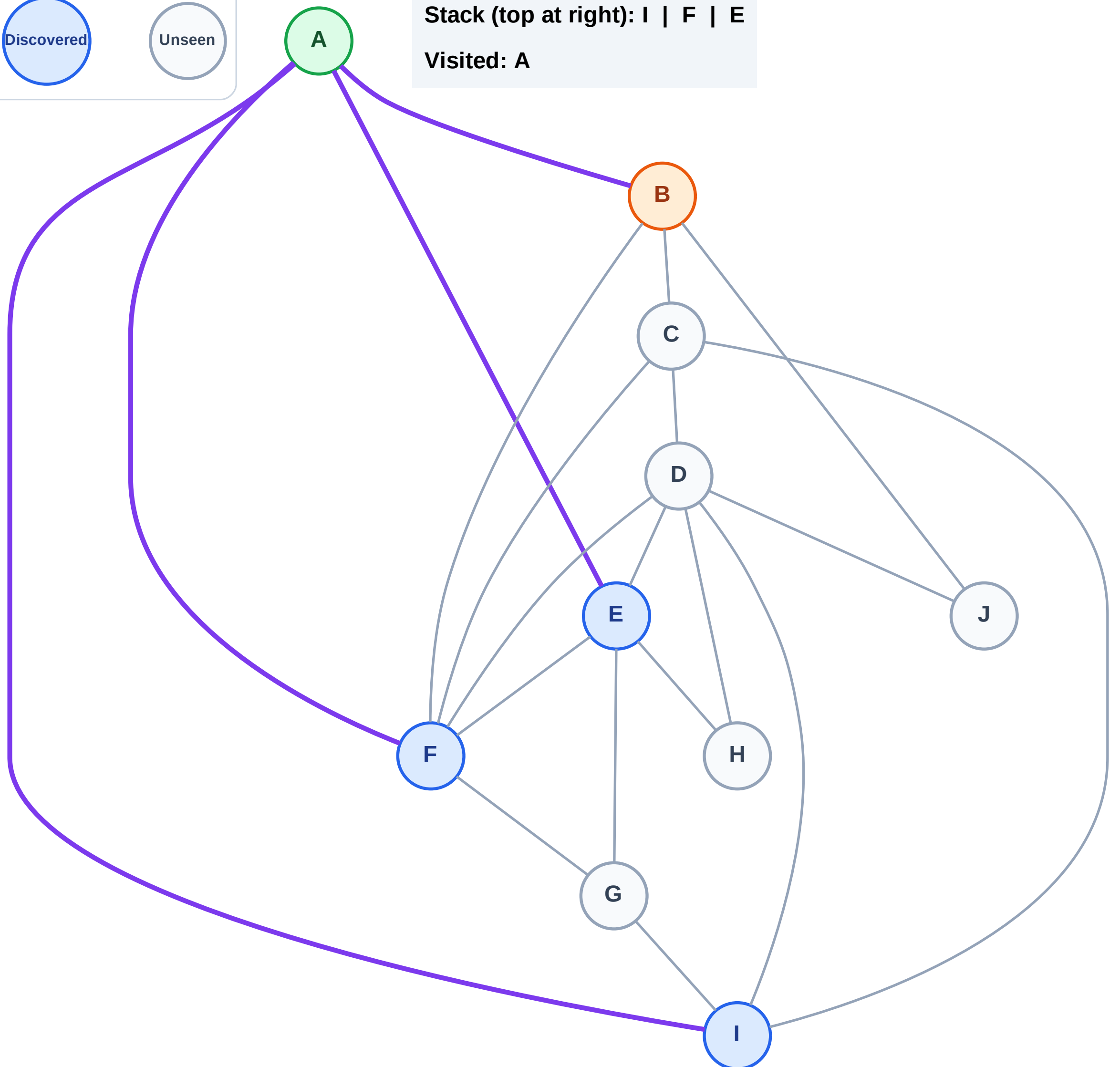
Step 4: Process node B

Legend



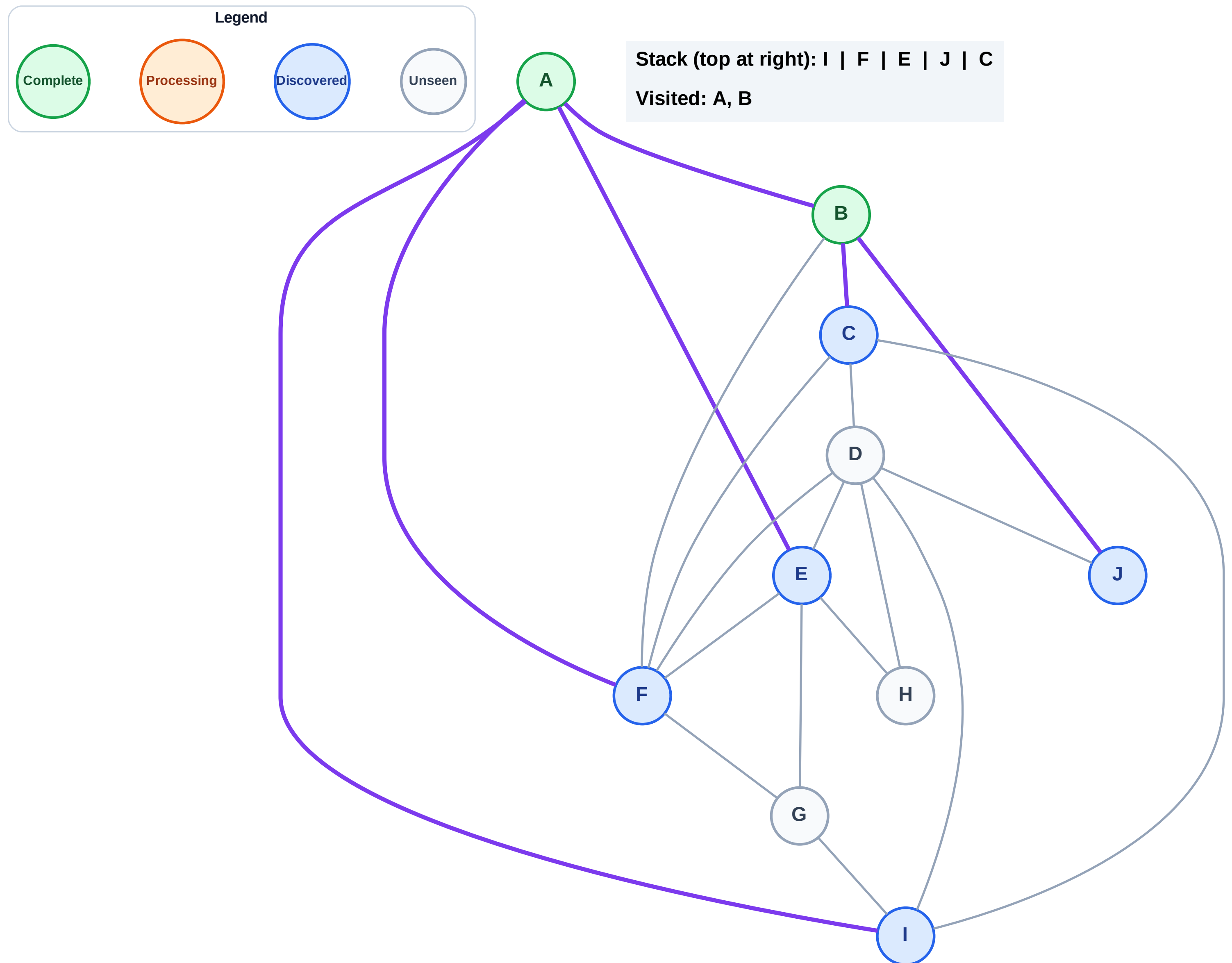
Stack (top at right): I | F | E

Visited: A



Step 5: Discover J, C from B

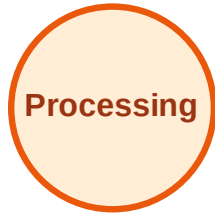
Step 5: Discover J, C from B



DFS Traversal

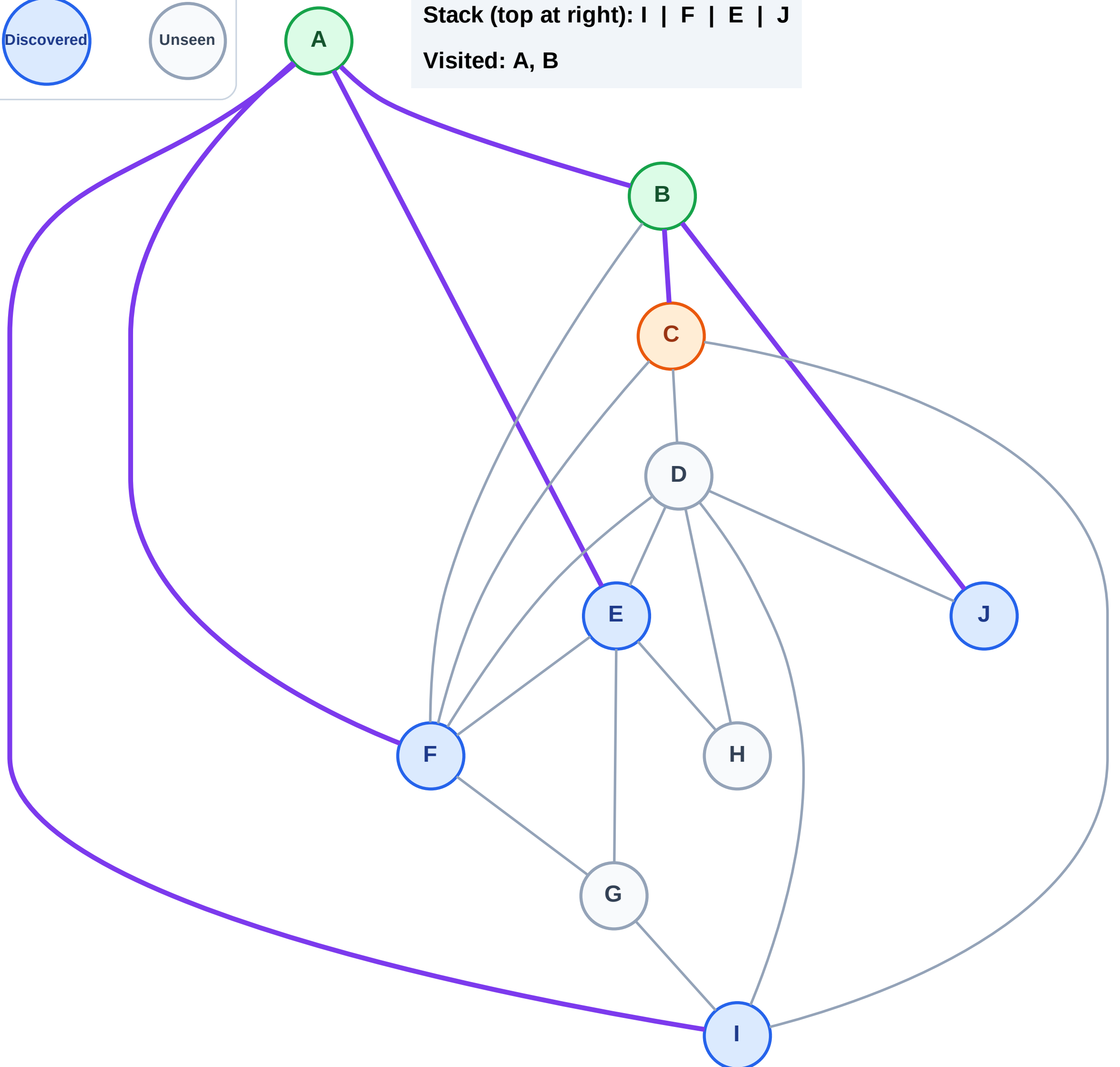
Step 6: Process node C

Legend



Stack (top at right): I | F | E | J

Visited: A, B



DFS Traversal

Step 7: Discover D from C

Legend

Complete

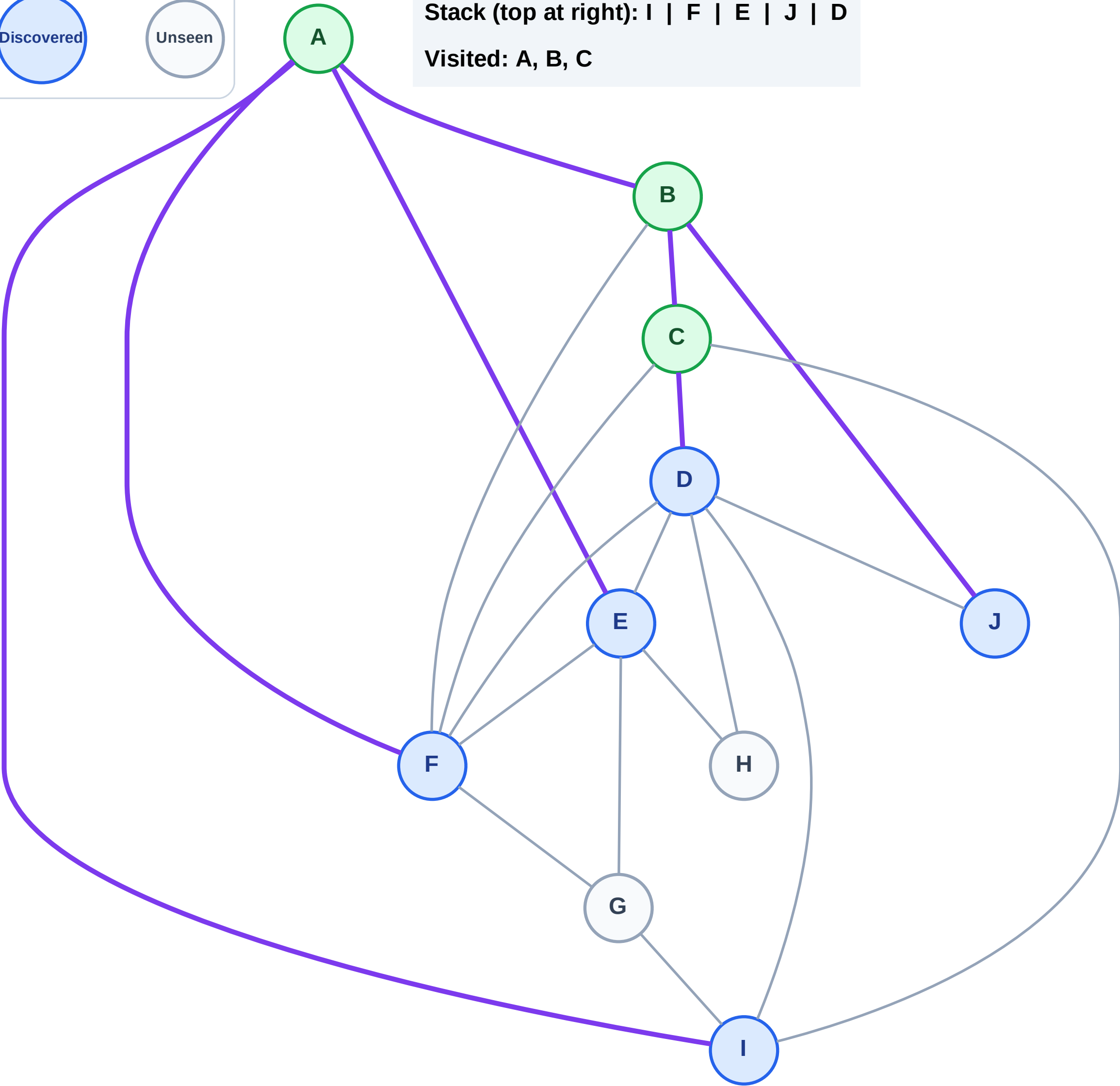
Processing

Discovered

Unseen

Stack (top at right): I | F | E | J | D

Visited: A, B, C



DFS Traversal

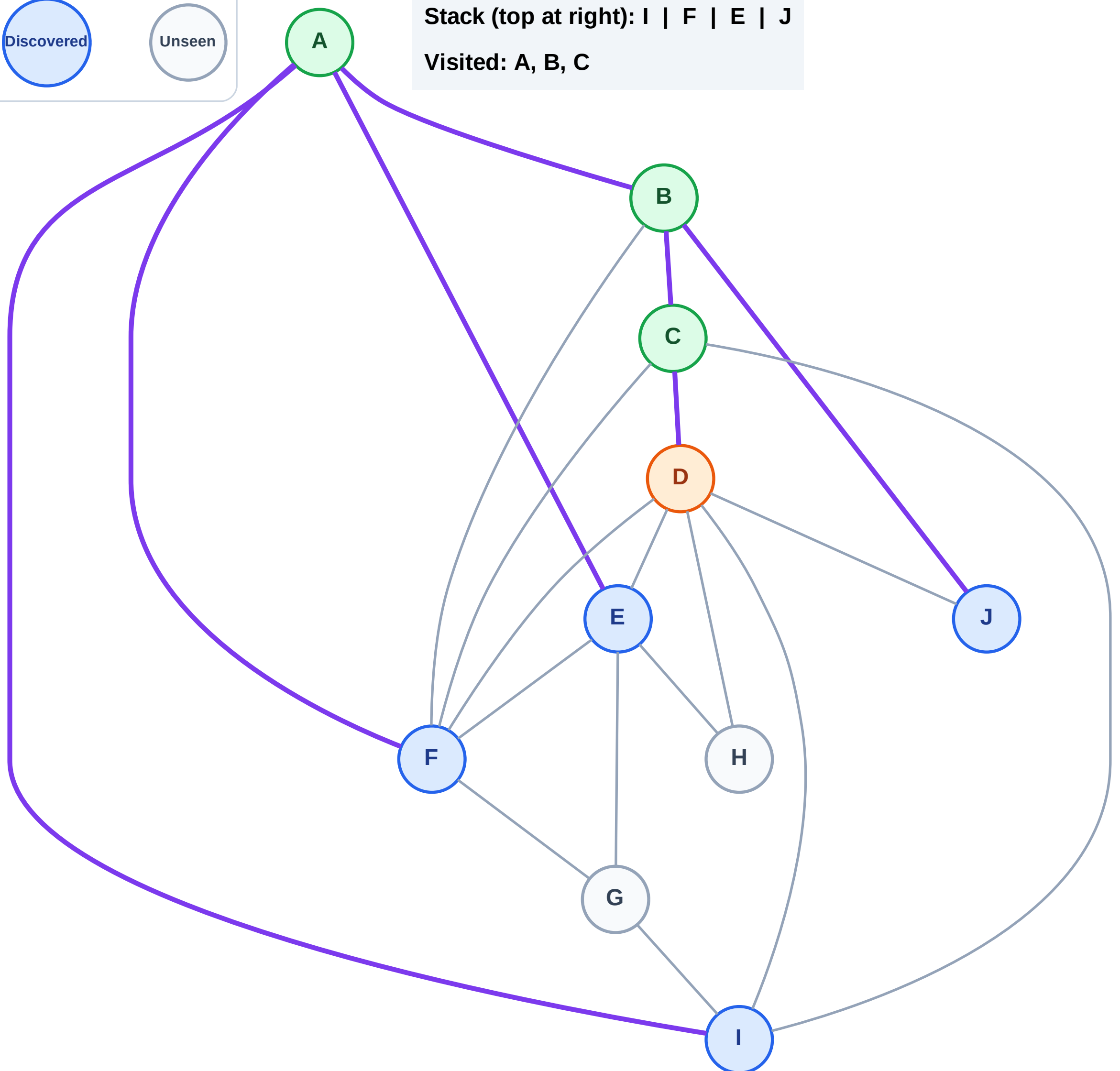
Step 8: Process node D

Legend

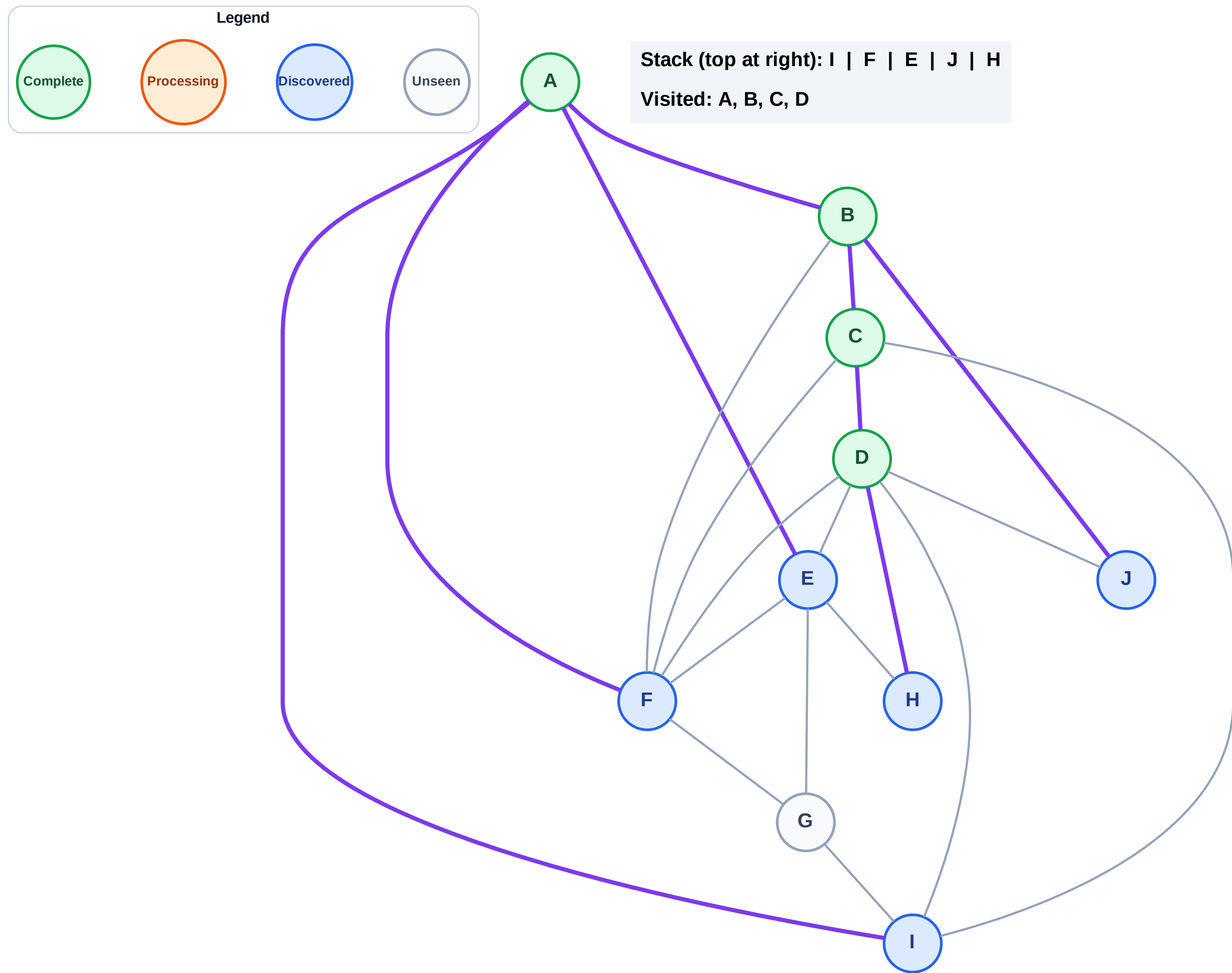


Stack (top at right): I | F | E | J

Visited: A, B, C



Step 9: Discover H from D



DFS Traversal

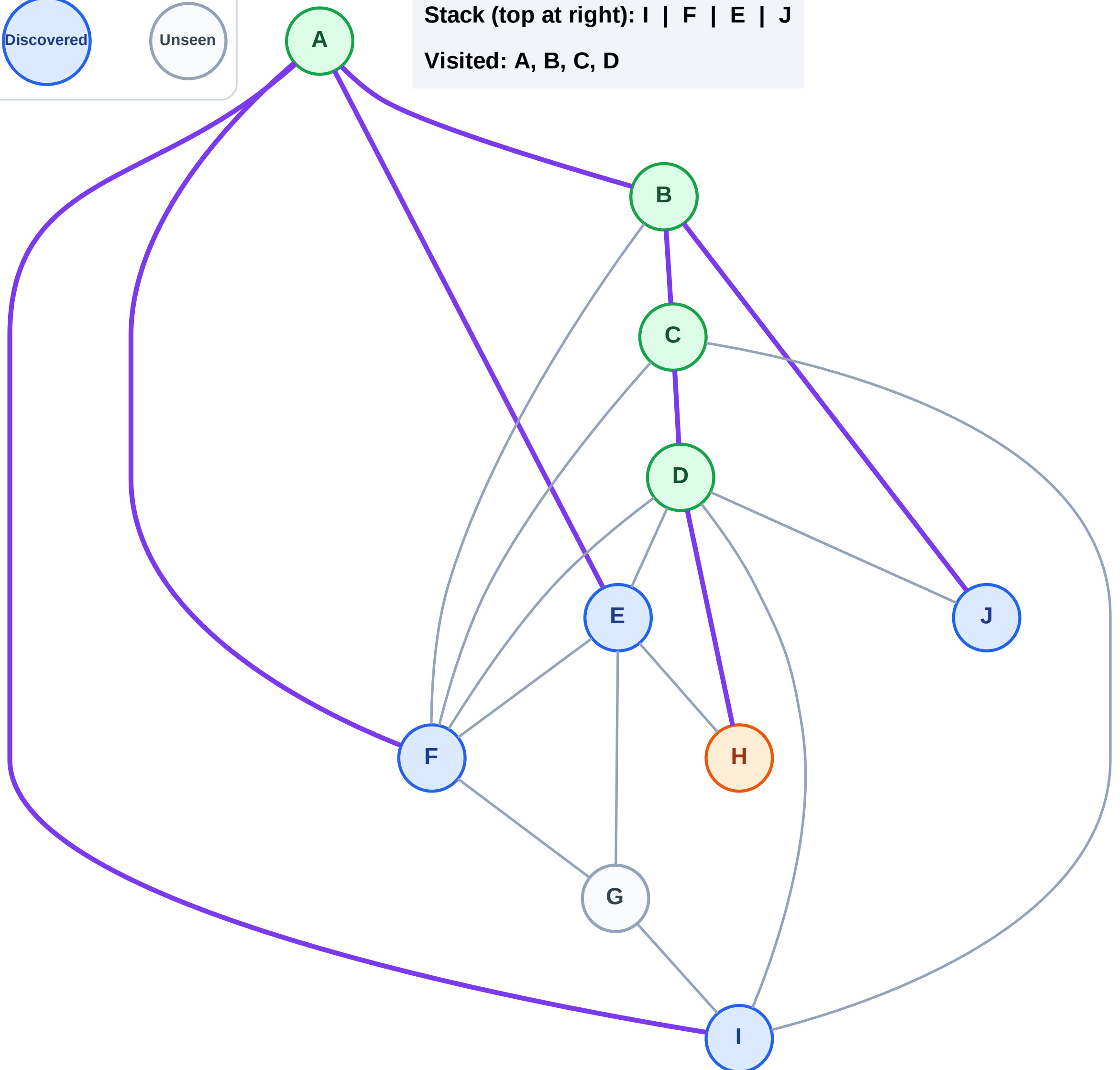
Step 10: Process node H

Legend



Stack (top at right): I | F | E | J

Visited: A, B, C, D



DFS Traversal

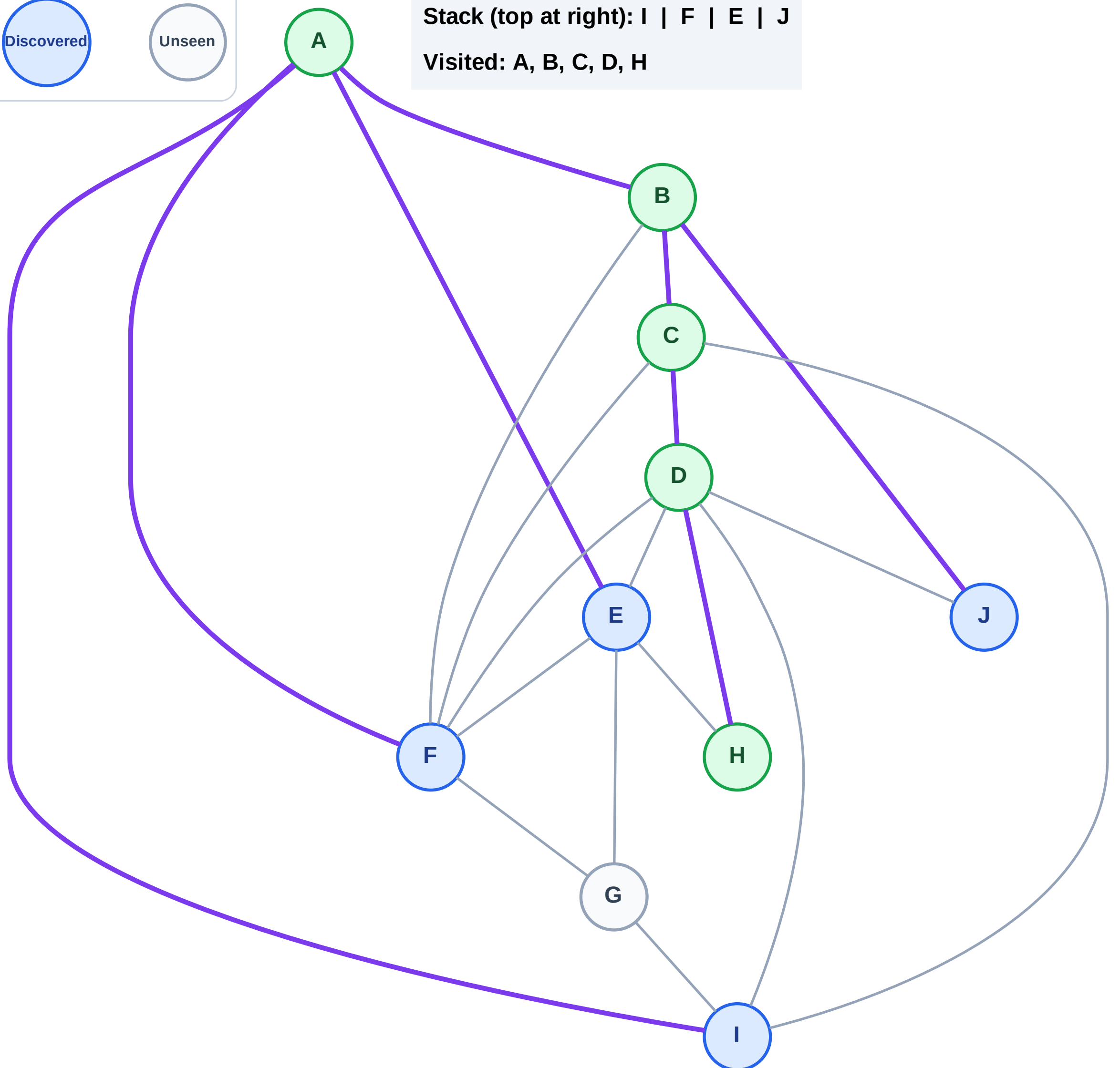
Step 11: No new neighbors from H

Legend



Stack (top at right): I | F | E | J

Visited: A, B, C, D, H



DFS Traversal

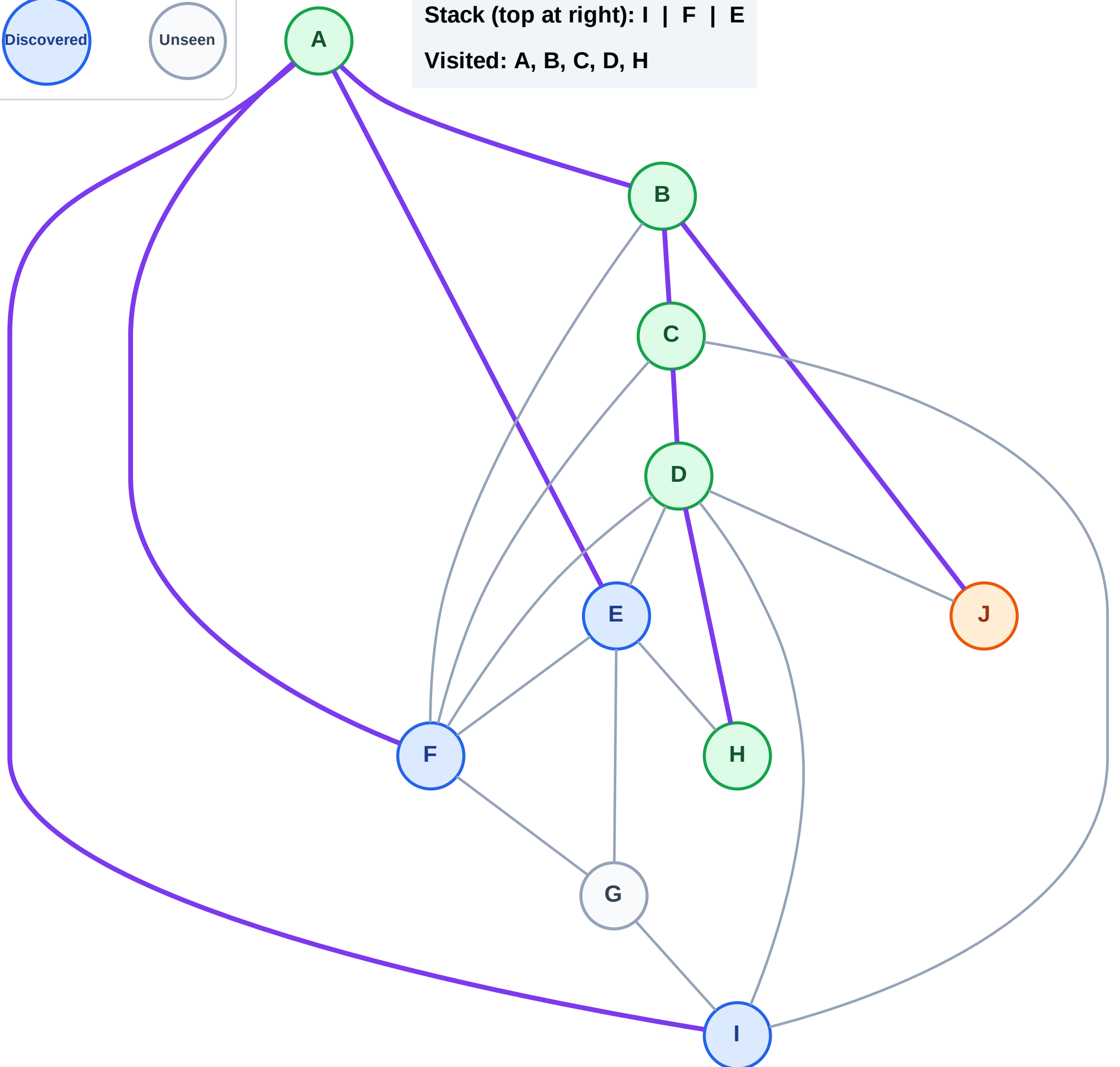
Step 12: Process node J

Legend



Stack (top at right): I | F | E

Visited: A, B, C, D, H



DFS Traversal

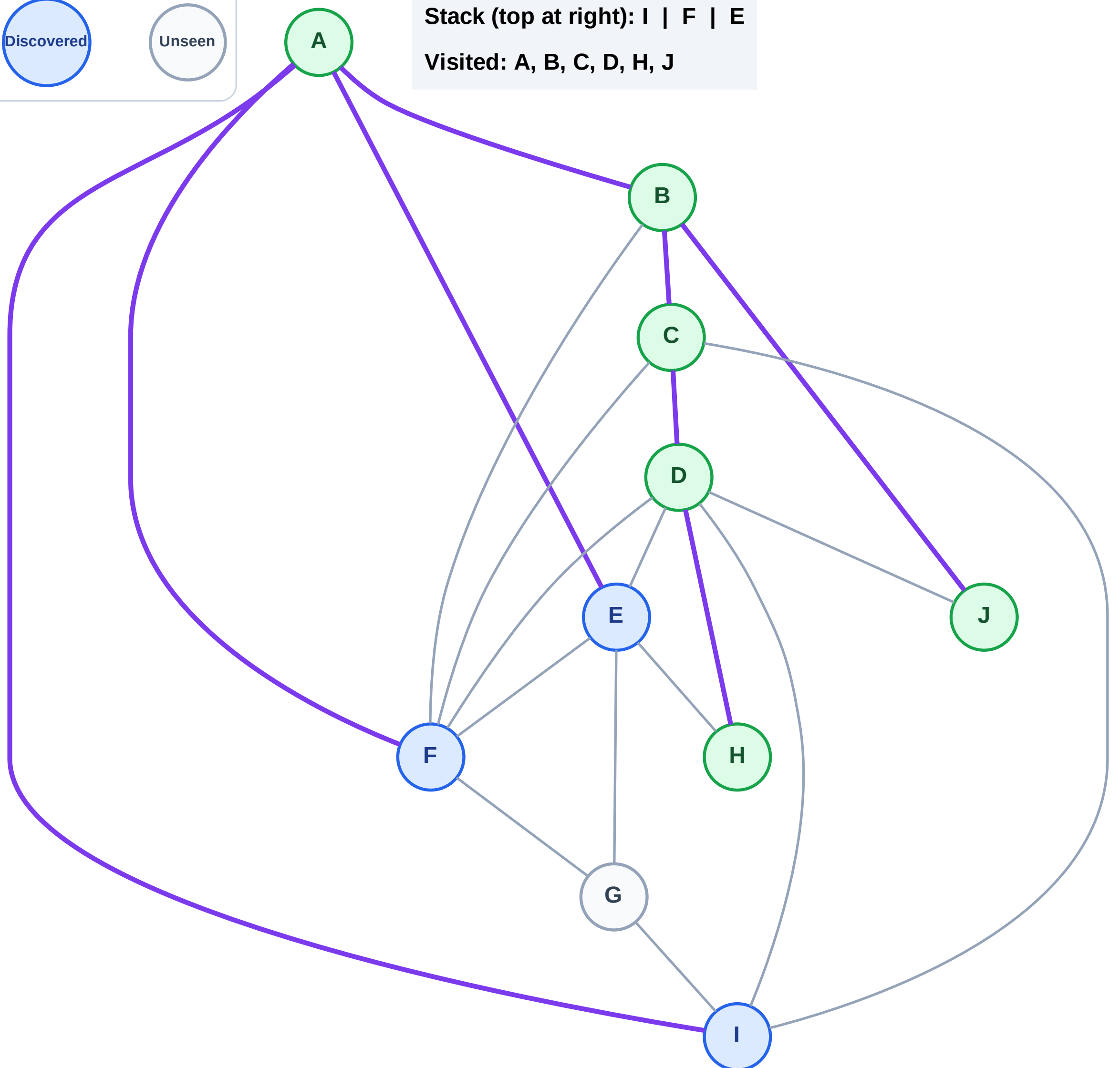
Step 13: No new neighbors from J

Legend



Stack (top at right): I | F | E

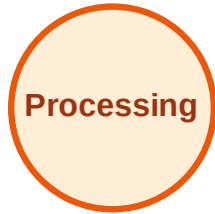
Visited: A, B, C, D, H, J



DFS Traversal

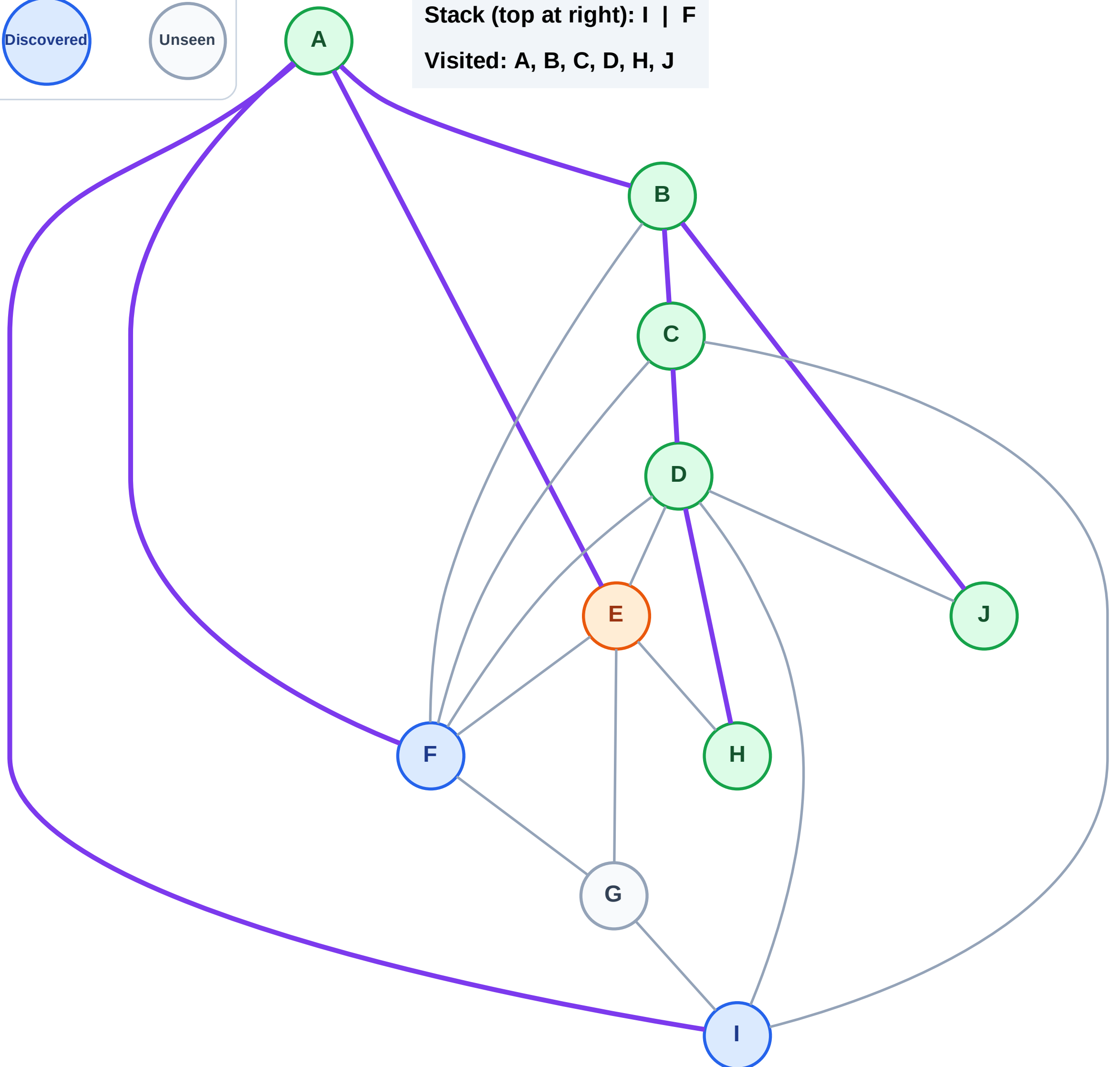
Step 14: Process node E

Legend



Stack (top at right): I | F

Visited: A, B, C, D, H, J



DFS Traversal

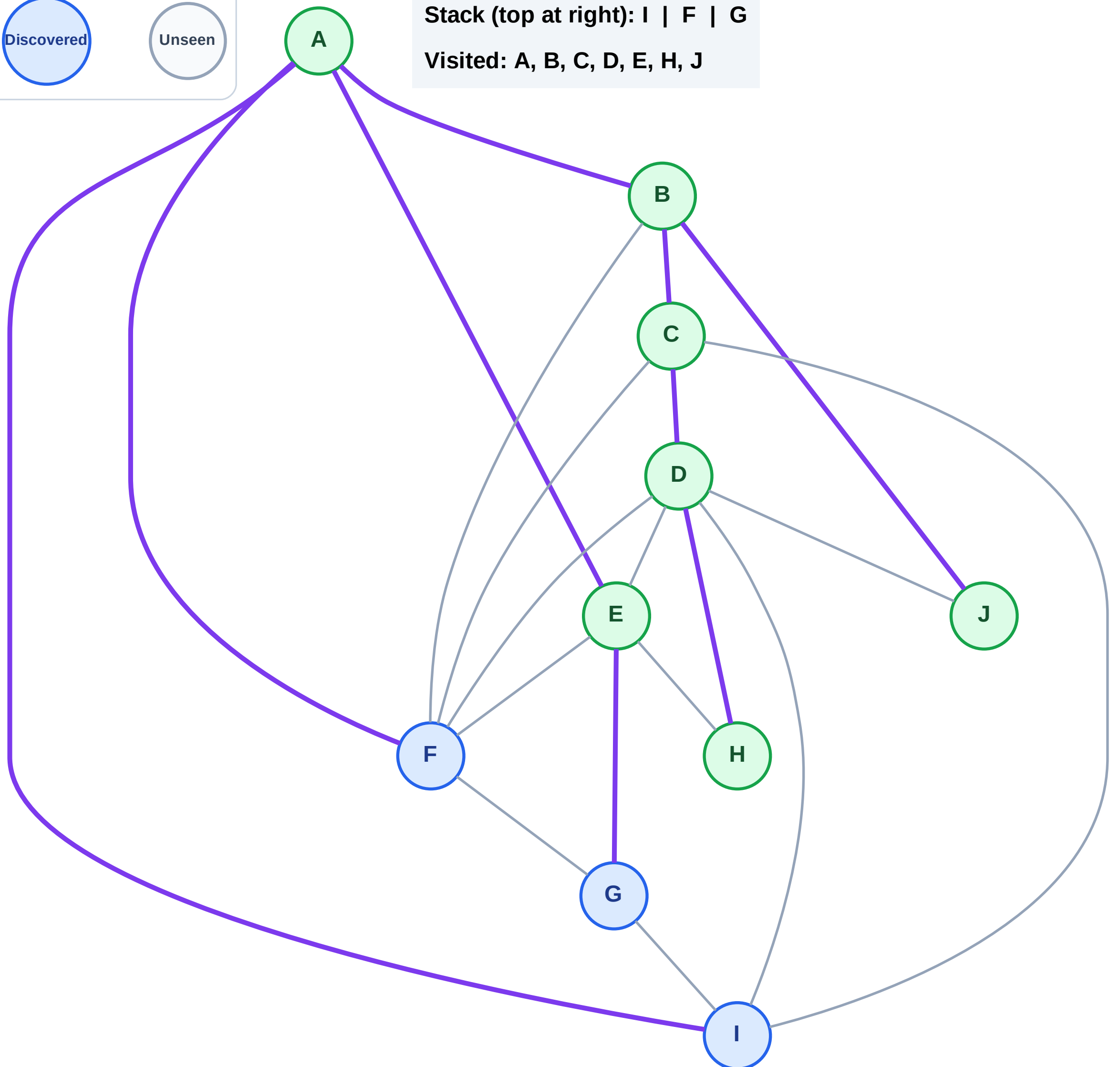
Step 15: Discover G from E

Legend



Stack (top at right): I | F | G

Visited: A, B, C, D, E, H, J



DFS Traversal

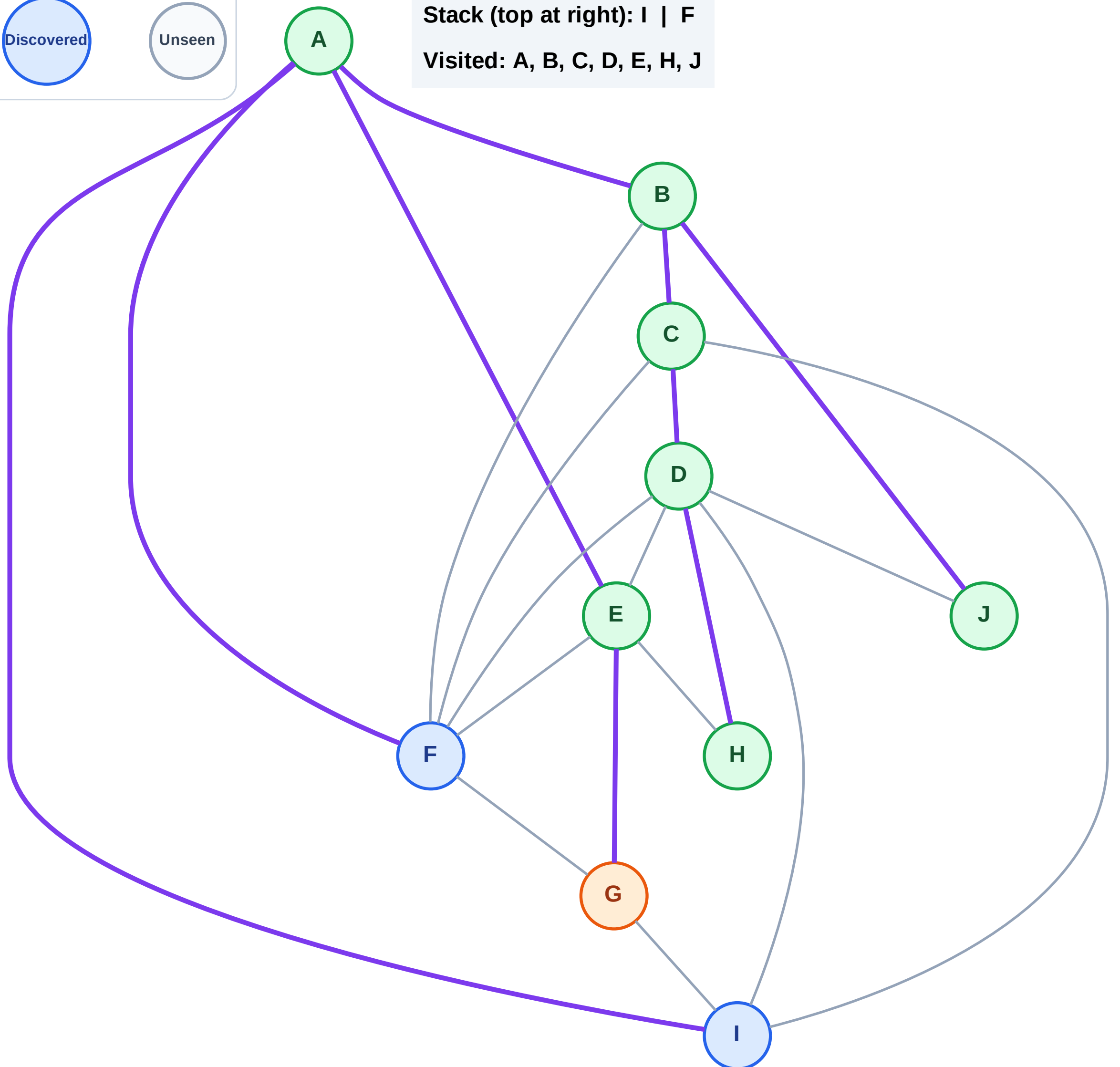
Step 16: Process node G

Legend



Stack (top at right): I | F

Visited: A, B, C, D, E, H, J



DFS Traversal

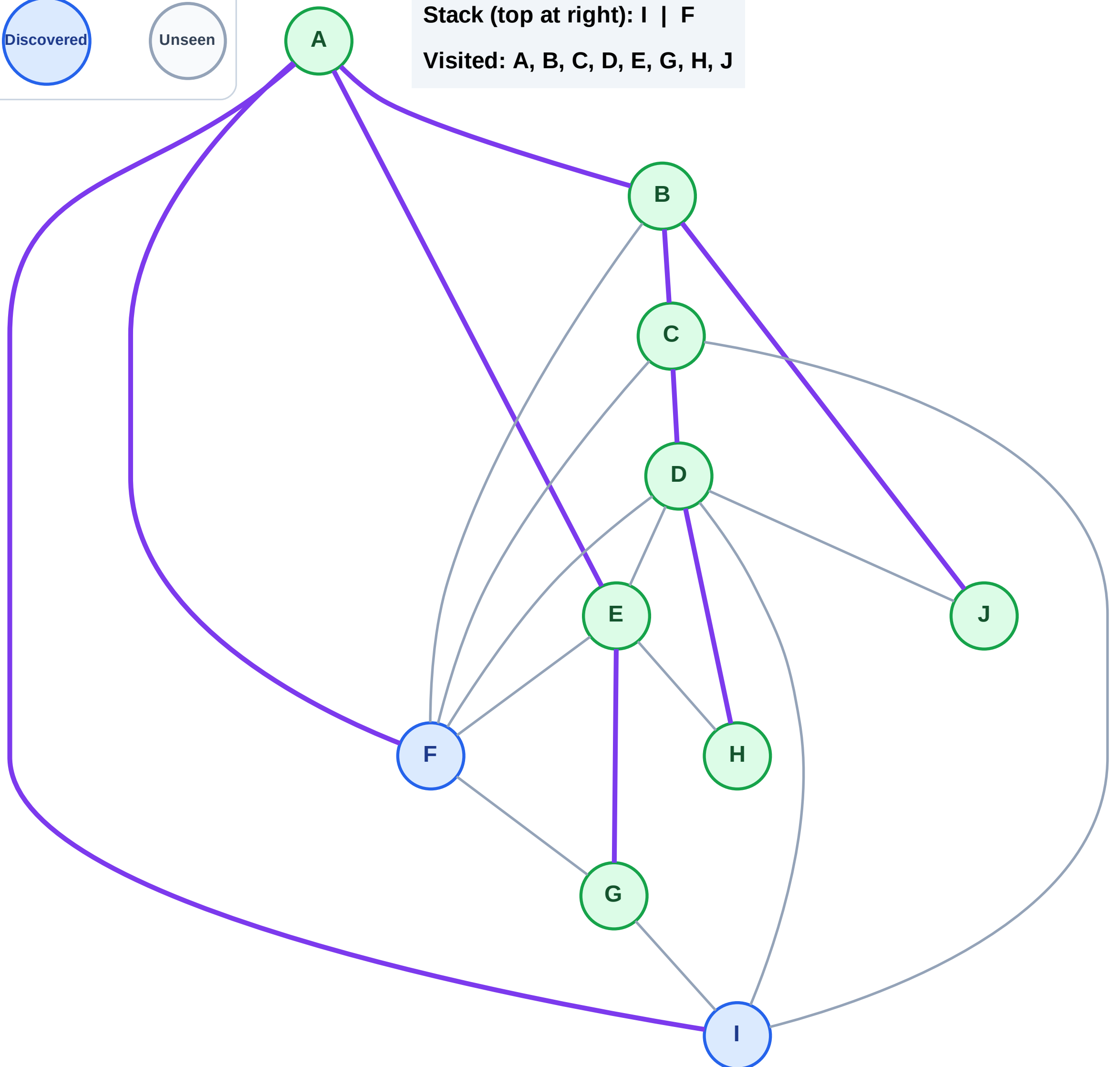
Step 17: No new neighbors from G

Legend



Stack (top at right): I | F

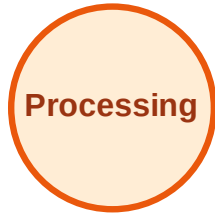
Visited: A, B, C, D, E, G, H, J



DFS Traversal

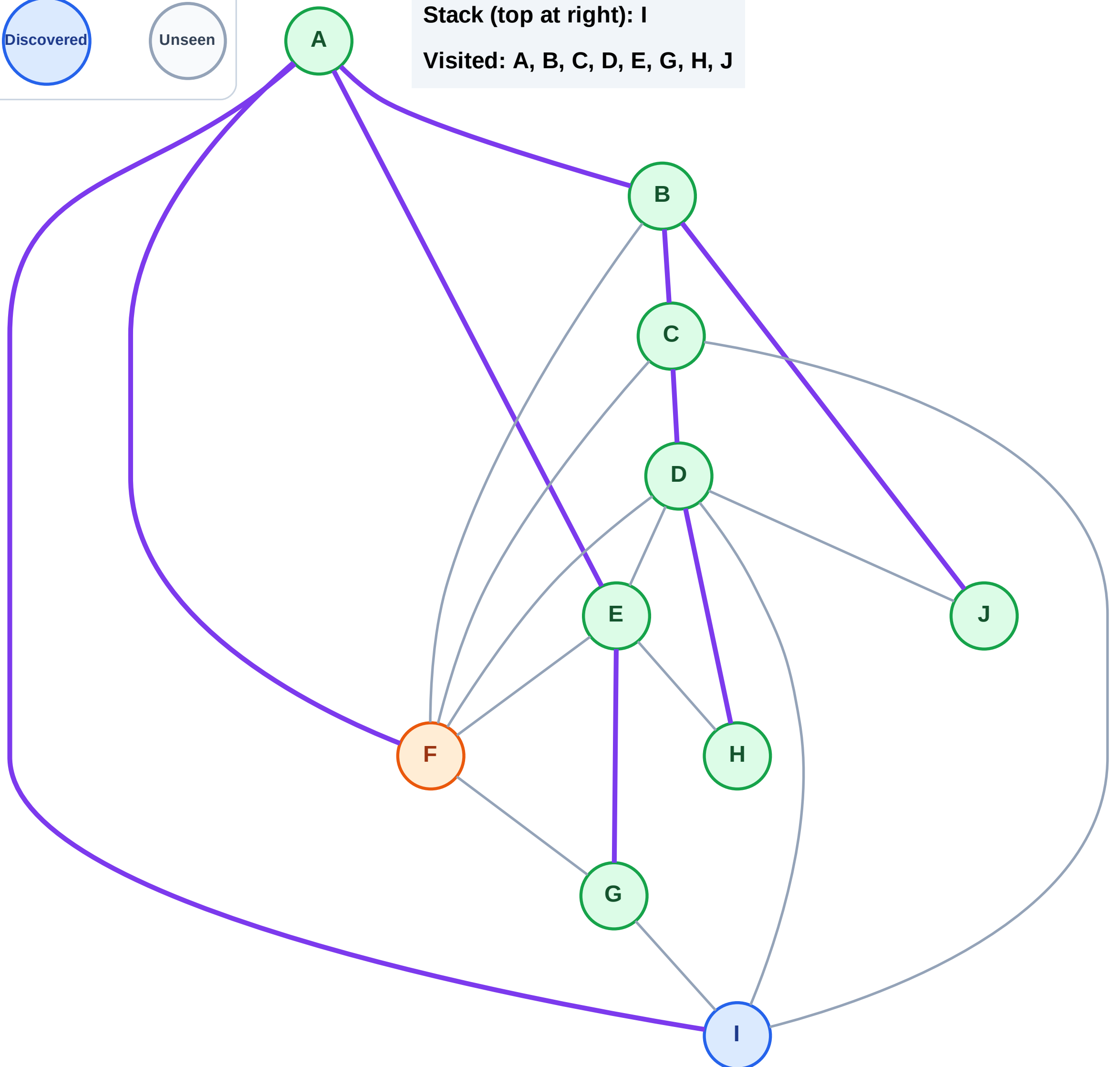
Step 18: Process node F

Legend



Stack (top at right): I

Visited: A, B, C, D, E, G, H, J



DFS Traversal

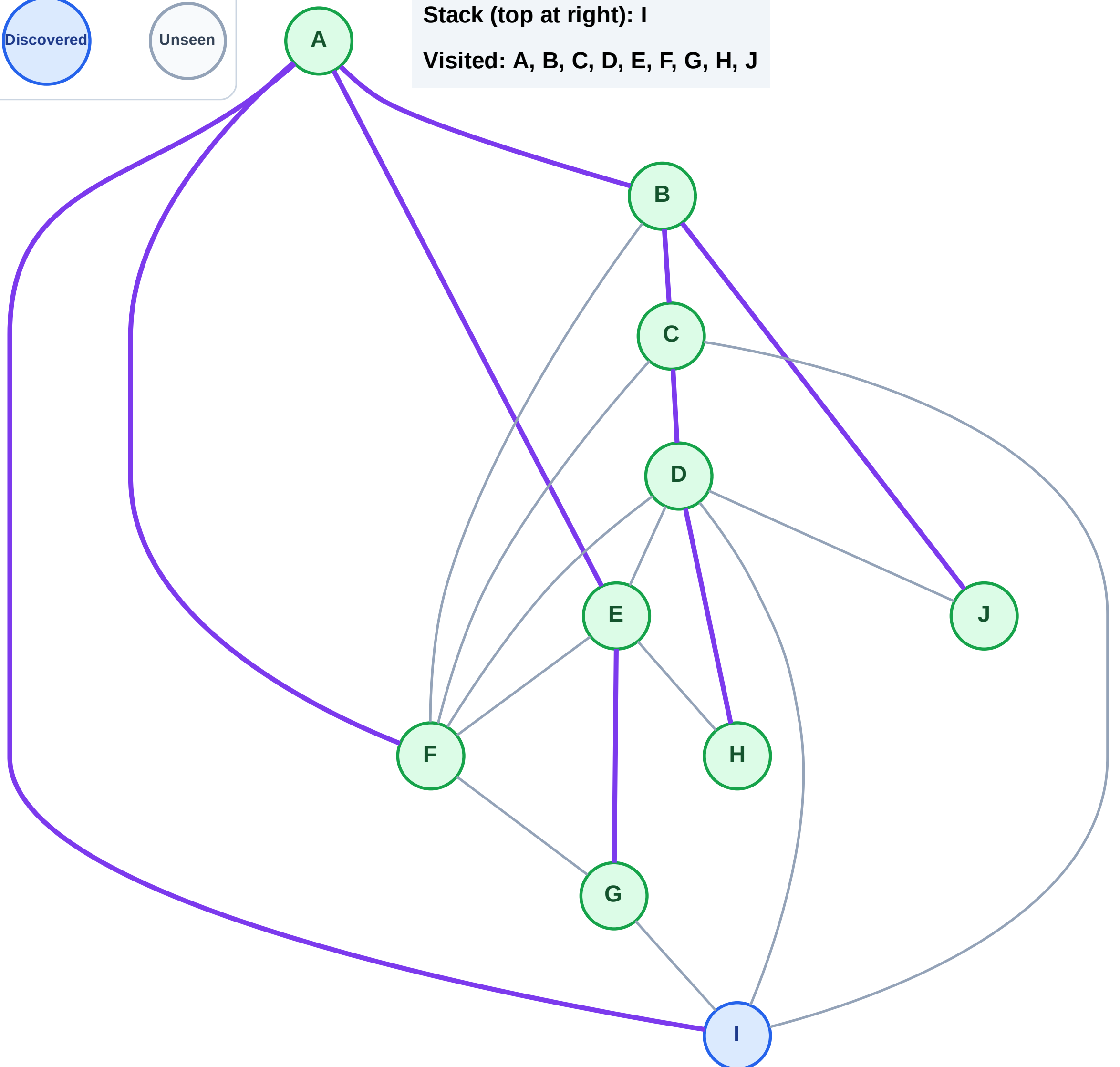
Step 19: No new neighbors from F

Legend



Stack (top at right): I

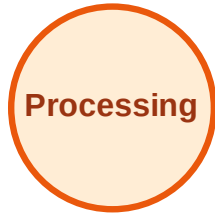
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

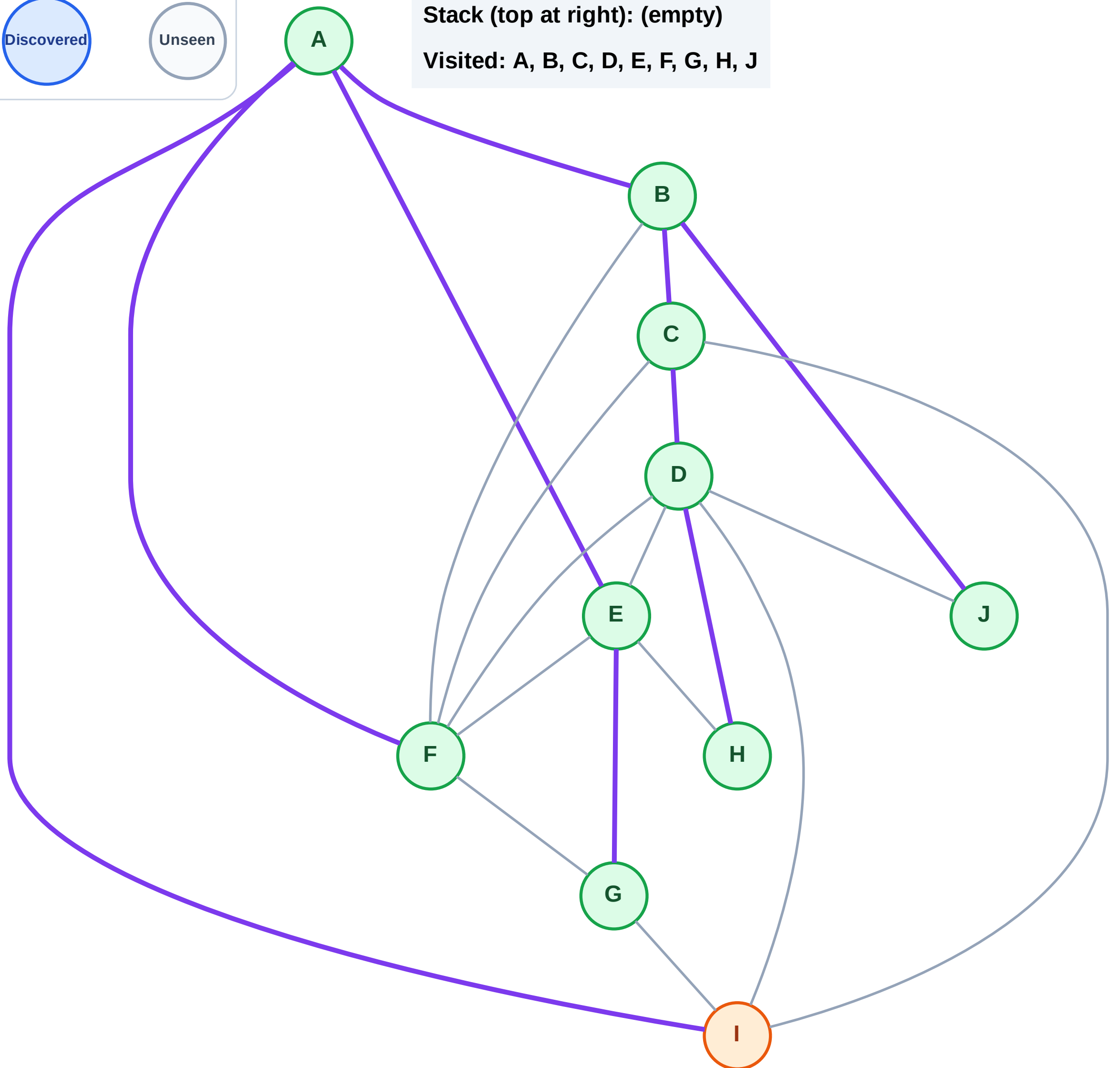
Step 20: Process node I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, J



Step 21: No new neighbors from I

Stack (top at right): (empty)

